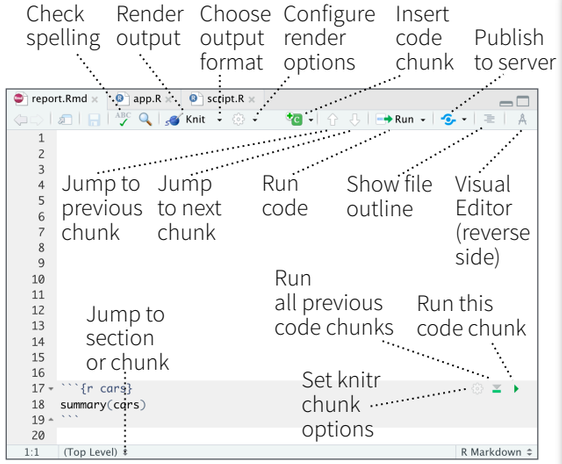
RStudio IDE : : **CHEATSHEET**

Documents and Apps

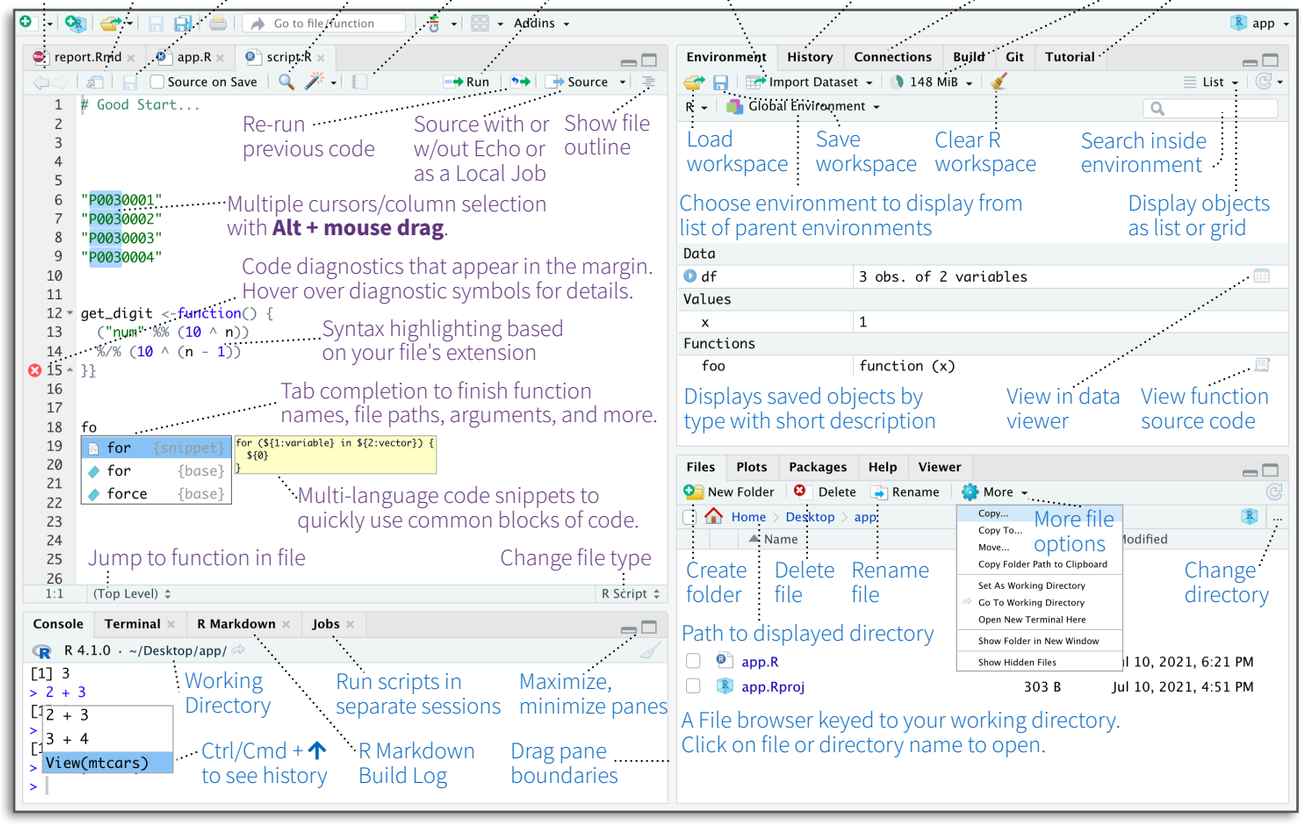
Open Shiny, R Markdown, knitr, Sweave, LaTeX, .Rd files and more in Source Pane



Access markdown guide at
Help > Markdown Quick Reference
See reverse side for more on **Visual Editor**

RStudio recognizes that files named **app.R**, **server.R**, **ui.R**, and **global.R** belong to a shiny app

Source Editor

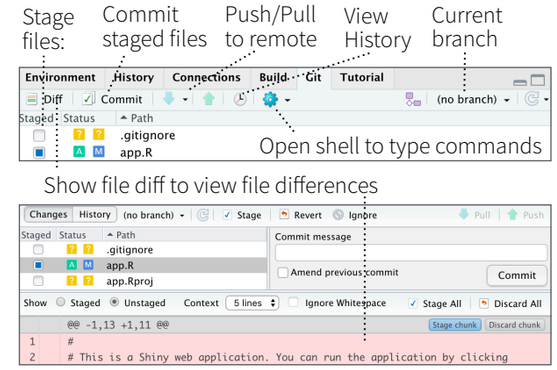


Tab Panes

Version Control

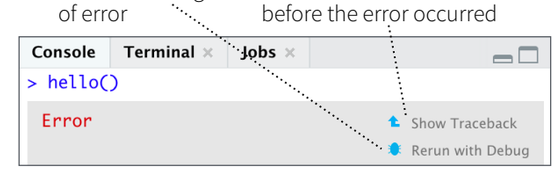
Turn on at **Tools > Project Options > Git/SVN**

A Added M Modified ? Untracked
D Deleted R Renamed



Debug Mode

Use **debug(), browser(),** or a breakpoint and execute your code to open the debugger mode.



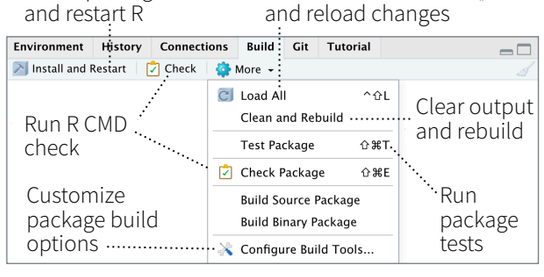
Package Development

Create a new package with
File > New Project > New Directory > R Package

Enable roxygen documentation with
Tools > Project Options > Build Tools

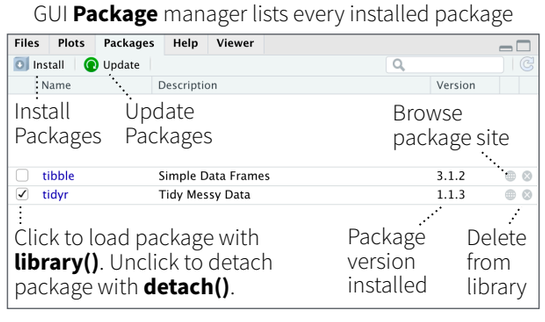
Roxygen guide at **Help > Roxygen Quick Reference**

See package information in the **Build Tab**

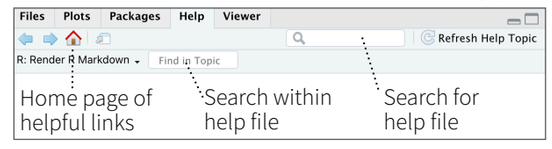


RStudio opens plots in a dedicated **Plots** pane

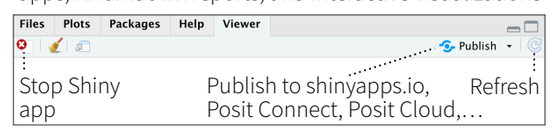
GUI **Package** manager lists every installed package



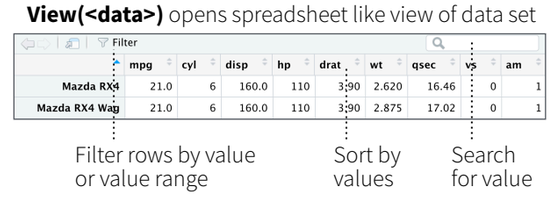
RStudio opens documentation in a dedicated **Help** pane



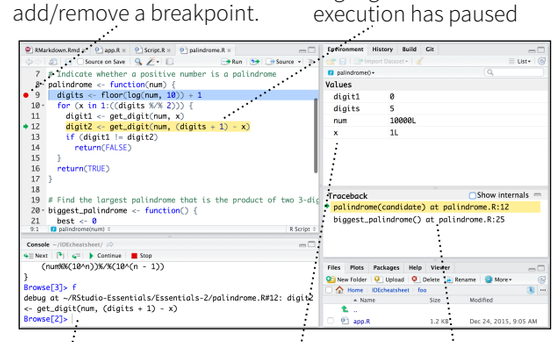
Viewer pane displays HTML content, such as Shiny apps, RMarkdown reports, and interactive visualizations



View(<data>) opens spreadsheet like view of data set



Click next to line number to add/remove a breakpoint.
Highlighted line shows where execution has paused



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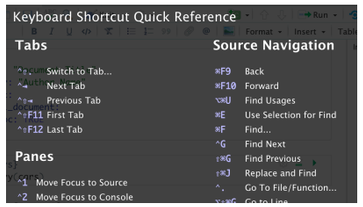
Keyboard Shortcuts

RUN CODE	Windows/Linux	Mac
Search command history	Ctrl+↑	Cmd+↑
Interrupt current command	Esc	Esc
Clear console	Ctrl+L	Ctrl+L
NAVIGATE CODE		
Go to File/Function	Ctrl+.	Ctrl+.
WRITE CODE		
Attempt completion	Tab or Ctrl+Space	Tab or Ctrl+Space
Insert <- (assignment operator)	Alt+-	Option+-
Insert |> or %>% (pipe operator)	Ctrl+Shift+M	Cmd+Shift+M
(Un)Comment selection	Ctrl+Shift+C	Cmd+Shift+C

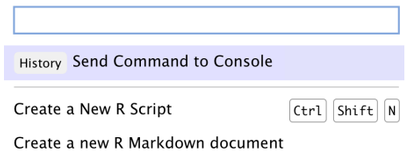
MAKE PACKAGES	Windows/Linux	Mac
Load All (devtools)	Ctrl+Shift+L	Cmd+Shift+L
Test Package (Desktop)	Ctrl+Shift+T	Cmd+Shift+T
Document Package	Ctrl+Shift+D	Cmd+Shift+D

DOCUMENTS AND APPS	Windows/Linux	Mac
Knit/Render Document (knitr)	Ctrl+Shift+K	Cmd+Shift+K
Insert chunk (Sweave & Knitr)	Ctrl+Alt+I	Cmd+Option+I
Run from start to current line	Ctrl+Alt+B	Cmd+Option+B
MORE KEYBOARD SHORTCUTS		
Keyboard Shortcuts Help	Alt+Shift+K	Option+Shift+K
Show Command Palette	Ctrl+Shift+P	Cmd+Shift+P

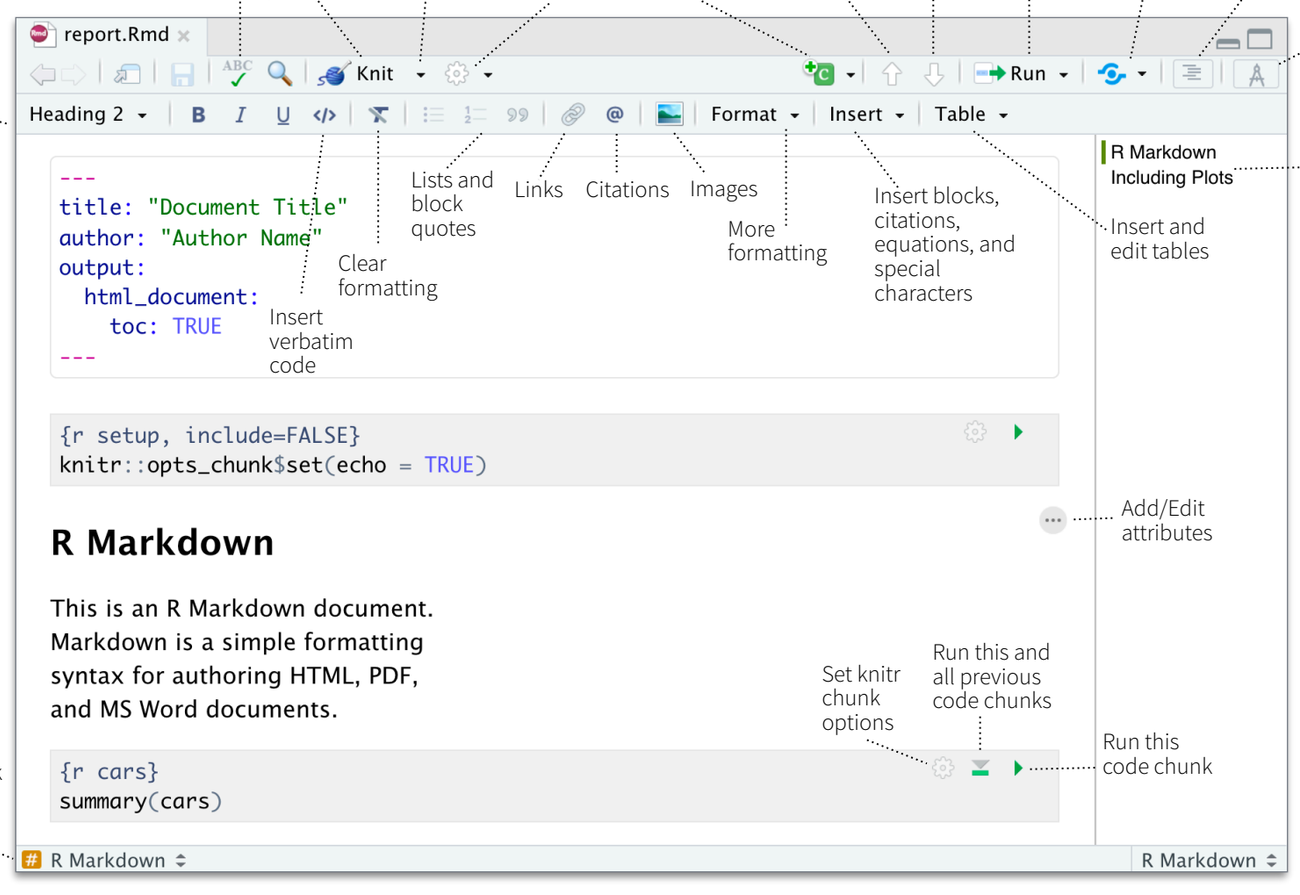
View the Keyboard Shortcut Quick Reference with **Tools > Keyboard Shortcuts** or **Alt/Option + Shift + K**



Search for keyboard shortcuts with **Tools > Show Command Palette** or **Ctrl/Cmd + Shift + P**.



Visual Editor



Posit Workbench

WHY POSIT WORKBENCH?

Extend the open source server with a commercial license, support, and more:

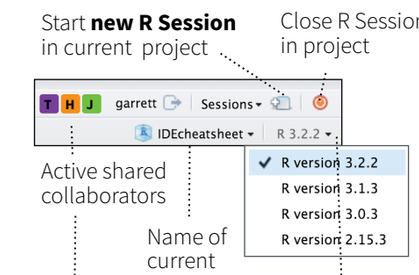
- open and run multiple R sessions at once
- tune your resources to improve performance
- administrative tools for managing user sessions
- collaborate real-time with others in shared projects
- switch easily from one version of R to a different version
- integrate with your authentication, authorization, and audit practices
- work in the RStudio IDE, JupyterLab, Jupyter Notebooks, or VS Code

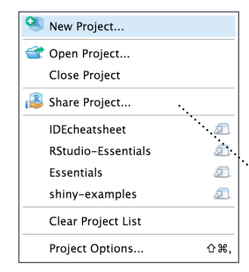
Download a free 45 day evaluation at **posit.co/products/enterprise/workbench/**

Share Projects

File > New Project

RStudio saves the call history, workspace, and working directory associated with a project. It reloads each when you re-open a project.

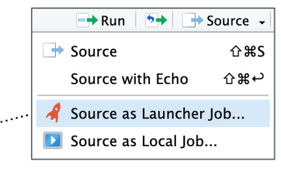




Share Project with Collaborators

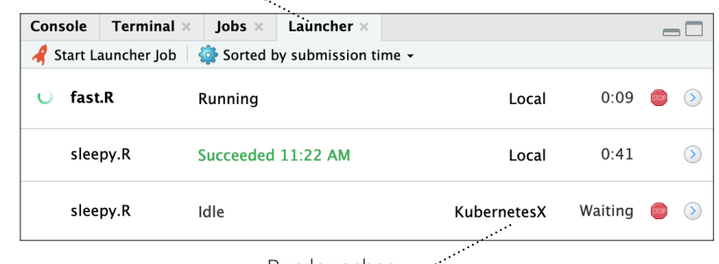
Run Remote Jobs

Run R on remote clusters (Kubernetes/Slurm) via the Job Launcher



Monitor launcher jobs

Launch a job



Run launcher jobs remotely

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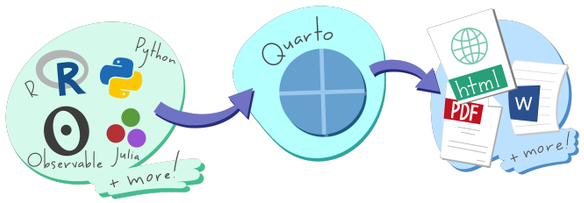
Publish and Share with Quarto : : **CHEATSHEET**



Author

WRITE AND CODE IN PLAIN TEXT

Author documents as .qmd files or Jupyter notebooks. Write in a rich Markdown syntax.

Render

GENERATE DOCUMENTS, PRESENTATIONS AND MORE

Produce HTML, PDF, MS Word reveal.js, MS Powerpoint, Beamer Websites, blogs, books...

Publish

SHARE YOUR WORK WITH THE WORLD

Quickly deploy to GitHub Pages, Netlify, Quarto Pub, Posit Cloud, or Posit Connect

GET QUARTO

https://quarto.org/docs/download/

Or use version **bundled with Positron or RStudio**

GET STARTED

https://quarto.org/docs/get-started/

Author

SOURCE FILE: hello.qmd

```
---
title: "Hello, Penguins"
format: html
execute:
  echo: false
---

## Meet the penguins

The `penguins` data contain
from three islands in the P

The three species of penguin
distributions of physical dimensions (@fig-penguins).

```{r}
#| label: fig-penguins
#| fig-cap: "Dimensions of penguins a
#| warning: false
library(tidyverse, quietly = TRUE)
library(palmerpenguins)
penguins |>
  ggplot(aes(x = flipper_length_mm, y = bill_length_mm)) +
  geom_point(aes(color = species)) +
  scale_color_manual(
    values = c("darkorange", "purple", "cyan4")) +
```
```

Set format(s) and options
Use YAML Syntax

Write with **Markdown**
RStudio: Help > Markdown Quick Reference
Use **Visual Editor**

Include code
R, Python, Julia, Observable, or any language with a Jupyter kernel

USE A TOOL WITH A RICH EDITING EXPERIENCE

RStudio, Positron, VS Code + Quarto extension

Run code cells as you write

Render with a button or keyboard shortcut

Edit Quarto documents with a **Visual Editor**

Apply formatting in Visual Editor. Saved as Markdown in source.

Insert elements like code cells, cross references, and more.

Normal | Format | Insert | Table

OR ANY TEXT EDITOR

Quarto documents (.qmd) can be edited in any tool that edits text.

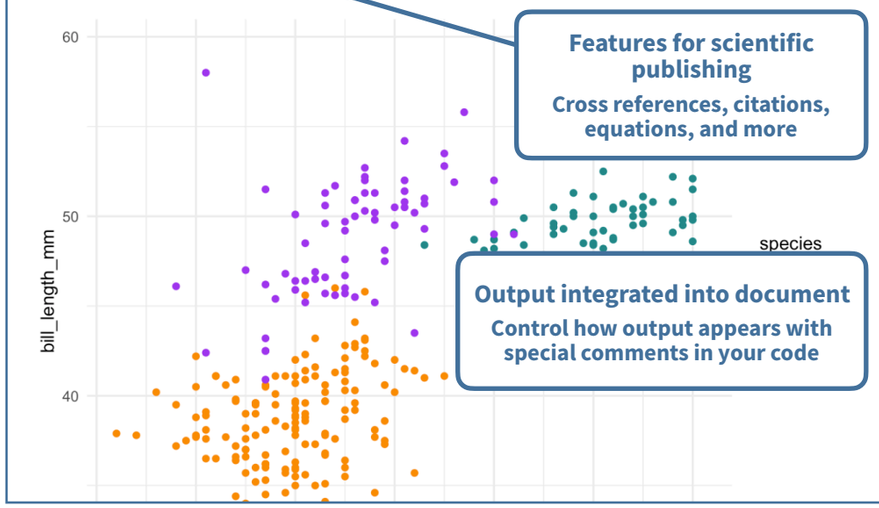
Render

RENDERED OUTPUT: hello.html

Hello, Penguins

Meet the penguins

The three species of penguins have quite distinct distributions of physical dimensions (Figure 1).



Features for scientific publishing
Cross references, citations, equations, and more

Output integrated into document
Control how output appears with special comments in your code

Save, then render to **preview** the document output.

```
Terminal
quarto preview hello.qmd
```

Use **Render** button

Use **Preview** button

The resulting HTML/PDF/MS Word/etc. document will be created and saved in the same directory as the source .qmd file.

BEHIND THE SCENES

When you render a document, Quarto:

1. Runs the code and embeds results and text into an .md file with: **Knitr**, if any {r} cells or, **Jupyter**, if any other cells.
2. Converts the .md file into the output format with Pandoc.

Publish

```
Terminal
quarto publish {venue} hello.qmd
```

{venue}: quarto-pub, connect, gh-pages, netlify, confluence

Quarto Pub — Free publishing service for Quarto content.

posit Connect — Org-hosted, control access, schedule updates.

Use **Publish** button

Use **Posit Publisher** extension

Quarto Projects

CREATE WEBSITES, BOOKS, AND MORE

A directory of Quarto documents + a configuration file (_quarto.yml)

See examples at https://quarto.org/docs/gallery/

Get started from the command line:

```
Terminal
quarto create project {type}
```

{type}: default, website, blog, book, confluence, manuscript

Use **File** > **New Project**

Use **Quarto: Create Project** command

Artwork from "Hello, Quarto" keynote by Julia Lowndes and Mine Çetinkaya-Rundel, presented at RStudio Conference 2022. Illustrated by Allison Horst.

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Include Code

CODE CELLS

Code cells start with ```{language} and end with ```.

Use **Insert Code Chunk/Cell**

```
```{r}
#| label: chunk-id
library(tidyverse)
```
```

```
```{python}
#| label: chunk-id
import pandas as pd
```
```

Other languages: `{julia}`, `{ojs}`

Add code cell options with `#|` comments.

Cell options control **execution**, figures, tables, layout and more. See them all at: https://quarto.org/docs/reference/cells

EXECUTION OPTIONS

OPTION	DEFAULT	EFFECTS
echo	true	false: hide code fenced: include code cell syntax
eval	true	false: don't run code
include	true	false: don't include code or results
output	true	false: don't include results asis: treat results as raw markdown
warning	true	false: don't include warnings in output
error	false	true: include error in output and continue with render

Set execution options at the **cell level**:

```
```{r}
#| echo: false
```
```

```
```{python}
#| echo: false
```
```

Or, **globally** in the YAML header with the **execute** option:

```
---
execute:
  echo: false
---
```

Set options in code cells with `#|` comments and YAML syntax:
`key: value`

INLINE CODE

Use computed values directly in text sections. Code is evaluated at render and results appear as text.

KNITR	JUPYTER	OUTPUT
Value is `r 2+2`.	Value is `{python} 2+2`.	Value is 4.

Set Format and Options

SET FORMAT OPTIONS

```
---
title: "My Document"
format:
  html:
    code-fold: true
    toc: true
---
```

Indent options 4 spaces

Indent format 2 spaces

MULTIPLE FORMATS

```
---
title: "My Document"
toc: true
format:
  html:
    code-fold: true
  pdf: default
---
```

Top-level options apply to all formats

Common formats: **html, pdf, docx, odt, rtf, gfm, pptx, revealjs, beamer**

Render **all** formats:

```
Terminal
quarto render hello.qmd
```

Render a **specific** format:

```
Terminal
quarto render hello.qmd --to pdf
```

	OPTION	html/revealjs	pdf/beamer	docx/pptx	DESCRIPTION
Nav	**toc**	X	X	X	Add a table of contents (true or false)
Nav	**toc-depth**	X	X	X	Lowest level of headings to add to table of contents (e.g. 2, 3)
Nav	**anchor-sections**	X			Show section anchors on mouse hover (true or false)
Style	**highlight-style**	X	X	X	Syntax highlighting theme (e.g. arrow, pygments, kate, zenburn)
Style	**mainfont, monofont**	X	X		Font name. HTML: sets CSS font-family; LaTeX: via fontspec package
Style	**theme**	X			Bootswatch theme name (e.g. cosmo, darkly, solar etc.)
Style	**css**	X			CSS or SCSS file to use to style the document (e.g. "style.css")
Style	**reference-doc**			X	docx/pptx file containing template styles (e.g. file.docx, file.pptx)
LaTeX	**include-in-header**	X	X		Files of content to include in header of output document, also **include-before-body, include-after-body**
LaTeX	**keep-md**	X	X	X	Keep intermediate markdown (true or false), also **keep-ipynb**, **keep-tex**
LaTeX	**documentclass**		X		LaTeX document class, set document class options with **classoption**
LaTeX	**pdf-engine**		X		LaTeX engine to produce PDF output (xelatex, pdflatex, lualatex)
LaTeX	**cite-method**		X		Method used to format citations (citeproc, natbib, biblatex)
Code	**code-fold**	X			Let readers toggle the display of R code (false, true, or show)
Code	**code-tools**	X			Add menu for hiding, showing, and downloading code (true or false)
Code	**code-overflow**	X			Display of wide code (scroll, or wrap)
Figures	**fig-align**	X	X	/	Alignment of figures (default, left, right, or center)
Figures	**fig-width, fig-height**	X	X	X	Default width and height for figures in inches
Figures	**fig-format**	X	X	X	Format for Matplotlib or R figures (retina, png, jpeg, svg, or pdf)

Also use in code cells ✓ ✓ ✓ Knitr

Visit https://quarto.org/docs/reference/ to see **all options** by format

Add Content

FIGURES

MARKDOWN

```
![CAP](image.png){#fig-LABEL fig-alt="ALT"}
```

COMPUTATION

```
```{python}
#| label: fig-LABEL
#| fig-cap: CAP
#| fig-alt: ALT
{{ plot code here }}
```
```

Or `{r}`

CROSS REFERENCES

1. **Add labels**
   Code cell: add option `label: prefix-LABEL`
   Markdown: add attribute `#prefix-LABEL`
2. **Add references** @prefix-LABEL, e.g.

```
You can see in @fig-scatterplot,
that...
```

Prefix	Renders	Prefix	Renders
fig-	Figure 1	eq-	Equation 1
tbl-	Table 1	sec-	Section 1

TABLES

MARKDOWN

```
|object | radius|
|:------|------:|
|Sun    | 696000|
|Earth  |   6371|

: CAPTION {#tbl-LABEL}
```

Insert Table in **Visual Editor**

COMPUTATION Output a Markdown table or an HTML table from your code

KNITR

Use knitr::kable() to produce Markdown:

```
```{r}
#| label: tbl-LABEL
#| tbl-cap: CAPTION
knitr::kable(head(cars))
```
```

Also see the R packages: gt, flextable, kableExtra.

JUPYTER Add `Markdown()` to Markdown output:

```
```{python}
#| label: tbl-LABEL
#| tbl-cap: CAPTION
import pandas as pd, tabulate
from IPython.display import Markdown
df = pd.DataFrame({"A": [1, 2],
                   "B": [1, 2]})
Markdown(df.to_markdown(index=False))
```
```

CITATIONS

1. Add a bibliography **file** to the YAML header:

```
---
bibliography: references.bib
---
```

2. Add citations: `[@citation]`, or `@citation`

Use **Insert Citations** dialog in the **Visual Editor**

Build your bibliography file from your Zotero library, DOI, Crossref, DataCite, or PubMed

CALLOUTS

tip

```
::: {.callout-tip}
## Title

Text
:::
```

Instead of **tip** use one of: note, caution, warning, or important.

note | warning | caution | important

SHORTCODES

```
{{< include _file.qmd >}}
{{< embed file.ipynb#id >}}
{{< video video.mp4 >}}
```

Data transformation with dplyr : : **CHEATSHEET**

dplyr functions work with pipes and expect **tidy data**. In tidy data:

Each **variable** is in its own **column** & Each **observation**, or **case**, is in its own **row**

pipes

x |> f(y) becomes **f(x, y)**

Summarize Cases

Apply **summary functions** to columns to create a new table of summary statistics. Summary functions take vectors as input and return one value (see back).

summary function

summarize(.data, …**)** Compute table of summaries.
mtcars |> summarize(avg = mean(mpg))

count(.data, …, wt = NULL, sort = FALSE, name = NULL**)** Count number of rows in each group defined by the variables in … Also **tally()**, **add_count()**, **add_tally()**.
mtcars |> count(cyl)

Group Cases

Use **group_by(**.data, …, .add = FALSE, .drop = TRUE**)** to create a "grouped" copy of a table grouped by columns in ... dplyr functions will manipulate each "group" separately and combine the results.

```
mtcars |>
  group_by(cyl) |>
  summarize(avg = mean(mpg))
```

Alternate grouping syntax with **.by** as an argument:

```
mtcars |>
 summarize(
   avg = mean(mpg), .by = cyl
)
```

Use **rowwise(**.data, …**)** to group data into individual rows. dplyr functions will compute results for each row. Also apply functions to list-columns. See tidyr cheat sheet for list-column workflow.

```
starwars |>
  rowwise() |>
  mutate(film_count = length(films))
```

ungroup(x, …**)** Returns ungrouped copy of table.

```
g_mtcars <- mtcars |> group_by(cyl)
ungroup(g_mtcars)
```

Manipulate Cases

EXTRACT CASES

Row functions return a subset of rows as a new table.

filter(.data, …, .preserve = FALSE**)** Extract rows that meet logical criteria.
mtcars |> filter(mpg > 20)

distinct(.data, …, .keep_all = FALSE**)** Remove rows with duplicate values.
mtcars |> distinct(gear)

slice(.data, …, .preserve = FALSE**)** Select rows by position.
mtcars |> slice(10:15)

slice_sample(.data, …, n, prop, weight_by = NULL, replace = FALSE**)** Randomly select rows. Use n to select a number of rows and prop to select a fraction of rows.
mtcars |> slice_sample(n = 5, replace = TRUE)

slice_min(.data, order_by, …, n, prop, with_ties = TRUE**)** and **slice_max()** Select rows with the lowest and highest values.
mtcars |> slice_min(mpg, prop = 0.25)

slice_head(.data, …, n, prop**)** and **slice_tail()** Select the first or last rows.
mtcars |> slice_head(n = 5)

Logical and boolean operators to use with filter()

==	<	<=	is.na()	%in%	|	xor()
!=	>	>=	!is.na()	!	&	

See **?base::Logic** and **?Comparison** for help.

ARRANGE CASES

arrange(.data, …, .by_group = FALSE**)** Order rows by values of a column or columns (low to high), use with **desc()** to order from high to low.
mtcars |> arrange(mpg)
mtcars |> arrange(desc(mpg))

ADD CASES

add_row(.data, …, .before = NULL, .after = NULL**)** Add one or more rows to a table.
cars |> add_row(speed = 1, dist = 1)

Manipulate Variables

EXTRACT VARIABLES

Column functions return a set of columns as a new vector or table.

pull(.data, var = -1, name = NULL, …**)** Extract column values as a vector, by name or index.
mtcars |> pull(wt)

select(.data, …**)** Extract columns as a table.
mtcars |> select(mpg, wt)

relocate(.data, …, .before = NULL, .after = NULL**)** Move columns to new position.
mtcars |> relocate(mpg, cyl, .after = last_col())

Use these helpers with select() and across()
e.g. mtcars |> select(mpg:cyl)

contains(match**)**	**num_range(**prefix, range**)**	**:**, e.g., mpg:cyl
ends_with(match**)**	**all_of(**x**)/any_of(**x, …, vars**)**	**!**, e.g., !gear
starts_with(match**)**	**matches(**match**)**	**everything()**

MANIPULATE MULTIPLE VARIABLES AT ONCE

```
df <- tibble(x_1 = c(1, 2), x_2 = c(3, 4), y = c(4, 5))
```

across(.cols, .funs, …, .names = NULL**)** Summarize or mutate multiple columns in the same way.
df |> summarize(across(everything(), mean))

c_across(.cols**)** Compute across columns in row-wise data.

```
df |>
 rowwise() |>
 mutate(x_total = sum(c_across(1:2)))
```

MAKE NEW VARIABLES

Apply **vectorized functions** to columns. Vectorized functions take vectors as input and return vectors of the same length as output (see back).

vectorized function

mutate(.data, …, .keep = "all", .before = NULL, .after = NULL**)** Compute new column(s). Also **add_column()**.
mtcars |> mutate(gpm = 1 / mpg)
mtcars |> mutate(gpm = 1 / mpg, .keep = "none")

rename(.data, …**)** Rename columns. Use **rename_with()** to rename with a function.
mtcars |> rename(miles_per_gallon = mpg)

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Vectorized Functions

TO USE WITH MUTATE ()

mutate() applies vectorized functions to columns to create new columns. Vectorized functions take vectors as input and return vectors of the same length as output.

vectorized function →

OFFSET

dplyr::**lag()** - offset elements by 1
dplyr::**lead()** - offset elements by -1

CUMULATIVE AGGREGATE

dplyr::**cumall()** - cumulative all()
dplyr::**cumany()** - cumulative any()
cummax() - cumulative max()
dplyr::**cummean()** - cumulative mean()
cummin() - cumulative min()
cumprod() - cumulative prod()
cumsum() - cumulative sum()

RANKING

dplyr::**cume_dist()** - proportion of all values <=
dplyr::**dense_rank()** - rank w ties = min, no gaps
dplyr::**min_rank()** - rank with ties = min
dplyr::**ntile()** - bins into n bins
dplyr::**percent_rank()** - min_rank scaled to [0,1]
dplyr::**row_number()** - rank with ties = "first"

MATH

+, - , *, /, ^, %/%, %% - arithmetic ops
log(), log2(), log10() - logs
<, <=, >, >=, !=, == - logical comparisons
dplyr::**between()** - x >= left & x <= right
dplyr::**near()** - safe == for floating point numbers

MISCELLANEOUS

dplyr::**case_when()** - multi-case if_else()

```
starwars |>
  mutate(type = case_when(
    height > 200 | mass > 200 ~ "large",
      species == "Droid"      ~ "robot",
      TRUE                    ~ "other")
    )
```

dplyr::**coalesce()** - first non-NA values by element across a set of vectors
dplyr::**if_else()** - element-wise if() + else()
dplyr::**na_if()** - replace specific values with NA
pmax() - element-wise max()
pmin() - element-wise min()

Summary Functions

TO USE WITH SUMMARIZE ()

summarize() applies summary functions to columns to create a new table. Summary functions take vectors as input and return single values as output.

summary function →

COUNT

dplyr::**n()** - number of values/rows
dplyr::**n_distinct()** - # of uniques
sum(!is.na()) - # of non-NAs

POSITION

mean() - mean, also **mean(!is.na())**
median() - median

LOGICAL

mean() - proportion of TRUEs
sum() - # of TRUEs

ORDER

dplyr::**first()** - first value
dplyr::**last()** - last value
dplyr::**nth()** - value in nth location of vector

RANK

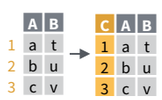
quantile() - nth quantile
min() - minimum value
max() - maximum value

SPREAD

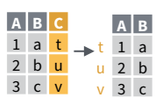
IQR() - Inter-Quartile Range
mad() - median absolute deviation
sd() - standard deviation
var() - variance

Row Names

Tidy data does not use rownames, which store a variable outside of the columns. To work with the rownames, first move them into a column.



tibble::**rownames_to_column()**
Move row names into col.
a <- mtcars |>
rownames_to_column(var = "C")

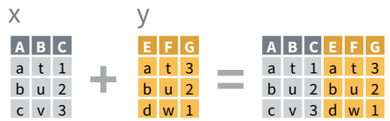


tibble::**column_to_rownames()**
Move col into row names.
a |> column_to_rownames(var = "C")

Also tibble::**has_rownames()** and tibble::**remove_rownames()**.

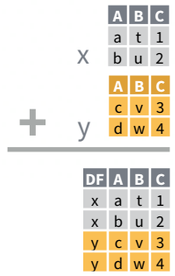
Combine Tables

COMBINE VARIABLES



bind_cols(…, .name_repair**)** Returns tables placed side by side as a single table. Column lengths must be equal. Columns will NOT be matched by id (to do that look at Relational Data below), so be sure to check that both tables are ordered the way you want before binding.

COMBINE CASES



bind_rows(…, .id = NULL**)** Returns tables one on top of the other as a single table. Set .id to a column name to add a column of the original table names (as pictured).

RELATIONAL DATA

Use a "**Mutating Join**" to join one table to columns from another, matching values with the rows that they correspond to. Each join retains a different combination of values from the tables.

left_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join matching values from y to x.

right_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join matching values from x to y.

inner_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join data. Retain only rows with matches.

full_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join data. Retain all values, all rows.

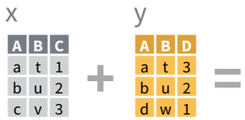
COLUMN MATCHING FOR JOINS

Use **by = c("col1", "col2", …)** to specify one or more common columns to match on.
left_join(x, y, by = "A")

Use a named vector, **by = c("col1" = "col2")**, to match on columns that have different names in each table.
left_join(x, y, by = c("C" = "D"))

Use **suffix** to specify the suffix to give to unmatched columns that have the same name in both tables.
left_join(x, y, by = c("C" = "D"), suffix = c("1", "2"))

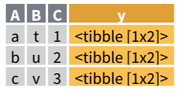
Use a "**Filtering Join**" to filter one table against the rows of another.



semi_join(x, y, by = NULL, copy = FALSE, …, na_matches = "na"**)** Return rows of x that have a match in y. Use to see what will be included in a join.

anti_join(x, y, by = NULL, copy = FALSE, …, na_matches = "na"**)** Return rows of x that do not have a match in y. Use to see what will not be included in a join.

Use a "**Nest Join**" to inner join one table to another into a nested data frame.



nest_join(x, y, by = NULL, copy = FALSE, keep = FALSE, name = NULL, …**)** Join data, nesting matches from y in a single new data frame column.

SET OPERATIONS

intersect(x, y, …)
Rows that appear in both x and y.

setdiff(x, y, …)
Rows that appear in x but not y.

union(x, y, …)
Rows that appear in x or y, duplicates removed). **union_all()** retains duplicates.

Use **setequal()** to test whether two data sets contain the exact same rows (in any order).

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Data visualization with ggplot2 : : CHEATSHEET

Basics

ggplot2 is based on the **grammar of graphics**, the idea that you can build every graph from the same components: a **data** set, a **coordinate system**, and **geoms**—visual marks that represent data points.

To display values, map variables in the data to visual properties of the geom (**aesthetics**) like **size**, **color**, and **x** and **y** locations.

Complete the template below to build a graph.

```
ggplot (data = <DATA>) +                    ] required
<GEOM_FUNCTION>(mapping = aes(<MAPPINGS>),
 stat =<STAT>, position =<POSITION>) +      ] Not required, sensible defaults supplied
<COORDINATE_FUNCTION> +
<FACET_FUNCTION> +
<SCALE_FUNCTION> +
<THEME_FUNCTION>
```

ggplot(data = mpg, **aes(**x = cty, y = hwy**))** Begins a plot that you finish by adding layers to. Add one geom function per layer.

last_plot() Returns the last plot.

ggsave("plot.png", width = 5, height = 5**)** Saves last plot as 5' x 5' file named "plot.png" in working directory. Matches file type to file extension.

Aes

Common aesthetic values.

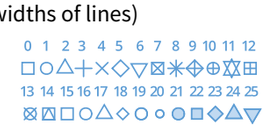
color and **fill** - string ("red", "#RRGGBB")

linetype - integer or string (0 = "blank", 1 = "solid", 2 = "dashed", 3 = "dotted", 4 = "dotdash", 5 = "longdash", 6 = "twodash")

size - integer (in mm for size of points and text)

linewidth - integer (in mm for widths of lines)

shape - integer/shape name or a single character ("a")



Geoms

Use a geom function to represent data points, use the geom's aesthetic properties to represent variables. Each function returns a layer.

GRAPHICAL PRIMITIVES

a <- ggplot(economics, aes(date, unemploy))
b <- ggplot(seals, aes(x = long, y = lat))

a + geom_blank() and **a + expand_limits()**
Ensure limits include values across all plots.

b + geom_curve(aes(yend = lat + 1, xend = long + 1), curvature = 1**)** - x, xend, y, yend, alpha, angle, color, curvature, linetype, size

a + geom_path(lineend = "butt", linejoin = "round", linemitre = 1**)**
x, y, alpha, color, group, linetype, size

a + geom_polygon(aes(alpha = 50)**)** - x, y, alpha, color, fill, group, subgroup, linetype, size

b + geom_rect(aes(xmin = long, ymin = lat, xmax = long + 1, ymax = lat + 1)**)** - xmax, xmin, ymax, ymin, alpha, color, fill, linetype, size

a + geom_ribbon(aes(ymin = unemploy - 900, ymax = unemploy + 900)**)** - x, ymax, ymin, alpha, color, fill, group, linetype, size

LINE SEGMENTS

common aesthetics: x, y, alpha, color, linetype, size

b + geom_abline(aes(intercept = 0, slope = 1)**)**
b + geom_hline(aes(yintercept = lat)**)**
b + geom_vline(aes(xintercept = long)**)**

b + geom_segment(aes(yend = lat + 1, xend = long + 1)**)**
b + geom_spoke(aes(angle = 1:1155, radius = 1)**)**

ONE VARIABLE continuous

c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)

c + geom_area(stat = "bin"**)**
x, y, alpha, color, fill, linetype, size

c + geom_density(kernel = "gaussian"**)**
x, y, alpha, color, fill, group, linetype, size, weight

c + geom_dotplot()
x, y, alpha, color, fill

c + geom_freqpoly()
x, y, alpha, color, group, linetype, size

c + geom_histogram(binwidth = 5**)**
x, y, alpha, color, fill, linetype, size, weight

c2 + geom_qq(aes(sample = hwy)**)**
x, y, alpha, color, fill, linetype, size, weight

discrete

d <- ggplot(mpg, aes(fl))

d + geom_bar()
x, alpha, color, fill, linetype, size, weight

TWO VARIABLES both continuous

e <- ggplot(mpg, aes(cty, hwy))

e + geom_label(aes(label = cty)**)** - x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

e + geom_point()
x, y, alpha, color, fill, shape, size, stroke

e + geom_quantile()
x, y, alpha, color, group, linetype, size, weight

e + geom_rug(sides = "bl"**)**
x, y, alpha, color, linetype, size

e + geom_smooth(method = lm**)**
x, y, alpha, color, fill, group, linetype, size, weight

e + geom_text(aes(label = cty)**)** - x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

one discrete, one continuous

f <- ggplot(mpg, aes(class, hwy))

f + geom_col()
x, y, alpha, color, fill, group, linetype, size

f + geom_boxplot()
x, y, lower, middle, upper, ymax, ymin, alpha, color, fill, group, linetype, shape, size, weight

f + geom_dotplot(binaxis = "y", stackdir = "center"**)**
x, y, alpha, color, fill, group

f + geom_violin(scale = "area"**)**
x, y, alpha, color, fill, group, linetype, size, weight

both discrete

g <- ggplot(diamonds, aes(cut, color))

g + geom_count()
x, y, alpha, color, fill, shape, size, stroke

e + geom_jitter(height = 2, width = 2**)**
x, y, alpha, color, fill, shape, size

continuous bivariate distribution

h <- ggplot(diamonds, aes(carat, price))

h + geom_bin2d(binwidth = c(0.25, 500)**)**
x, y, alpha, color, fill, linetype, size, weight

h + geom_density_2d()
x, y, alpha, color, group, linetype, size

h + geom_hex()
x, y, alpha, color, fill, size

continuous function

i <- ggplot(economics, aes(date, unemploy))

i + geom_area()
x, y, alpha, color, fill, linetype, size

i + geom_line()
x, y, alpha, color, group, linetype, size

i + geom_step(direction = "hv"**)**
x, y, alpha, color, group, linetype, size

visualizing error

df <- data.frame(grp = c("A", "B"), fit = 4:5, se = 1:2)
j <- ggplot(df, aes(grp, fit, ymin = fit - se, ymax = fit + se))

j + geom_crossbar(fatten = 2**)** - x, y, ymax, ymin, alpha, color, fill, group, linetype, size

j + geom_errorbar() - x, ymax, ymin, alpha, color, group, linetype, size, width
Also **geom_errorbarh()**.

j + geom_linerange()
x, ymin, ymax, alpha, color, group, linetype, size

j + geom_pointrange() - x, y, ymin, ymax, alpha, color, fill, group, linetype, shape, size

maps

Draw the appropriate geometric object depending on the simple features present in the data. aes() arguments: map_id, alpha, color, fill, linetype, linewidth.

nc <- **sf::st_read(**system.file("shape/nc.shp", package = "sf"**)**)

ggplot(nc) +
geom_sf(aes(fill = AREA)**)**

THREE VARIABLES

seals$z <- with(seals, sqrt(delta_long^2 + delta_lat^2)); l <- ggplot(seals, aes(long, lat))

l + geom_contour(aes(z = z)**)**
x, y, z, alpha, color, group, linetype, size, weight

l + geom_contour_filled(aes(fill = z)**)**
x, y, alpha, color, fill, group, linetype, size, subgroup

l + geom_raster(aes(fill = z), hjust = 0.5, vjust = 0.5, interpolate = FALSE**)**
x, y, alpha, fill

l + geom_tile(aes(fill = z)**)**
x, y, alpha, color, fill, linetype, size, width

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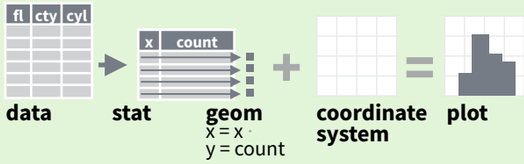
Stats An alternative way to build a layer.

A stat builds new variables to plot (e.g., count, prop).



Visualize a stat by changing the default stat of a geom function, **geom_bar(stat="count")** or by using a stat function, **stat_count(geom="bar")**, which calls a default geom to make a layer (equivalent to a geom function).
Use **after_stat(name)** syntax to map the stat variable **name** to an aesthetic.

i + stat_density_2d(aes(fill = after_stat(level)), geom = "polygon"**)**

geom to use | stat function | geommappings | variable created by stat

c + stat_bin(binwidth = 1, boundary = 10**)**
x, y | count, ncount, density, ndensity

c + stat_count(width = 1**) x, y** | count, prop

c + stat_density(adjust = 1, kernel = "gaussian"**)**
x, y | count, density, scaled

e + stat_bin_2d(bins = 30, drop = T**)**
x, y, fill | count, density

e + stat_bin_hex(bins = 30**) x, y, fill** | count, density

e + stat_density_2d(contour = TRUE, n = 100**)**
x, y, color, size | level

e + stat_ellipse(level = 0.95, segments = 51, type = "t"**)**

l + stat_contour(aes(z = z)**) x, y, z, order** | level

l + stat_summary_hex(aes(z = z), bins = 30, fun = max**)**
x, y, z, fill | value

l + stat_summary_2d(aes(z = z), bins = 30, fun = mean**)**
x, y, z, fill | value

f + stat_boxplot(coef = 1.5**)**
x, y | lower, middle, upper, width , ymin, ymax

f + stat_ydensity(kernel = "gaussian", scale = "area"**) x, y** | density, scaled, count, n, violinwidth, width

e + stat_ecdf(n = 40**) x, y** | x, y

e + stat_quantile(quantiles = c(0.1, 0.9), formula = y ~ log(x), method = "rq"**) x, y** | quantile

e + stat_smooth(method = "lm", formula = y ~ x, se = T, level = 0.95**) x, y** | se, x, y, ymin, ymax

ggplot() + xlim(-5, 5**) + stat_function(**fun = dnorm, n = 20, geom = "point"**) x** | x, y

ggplot() + stat_qq(aes(sample = 1:100)**)**
x, y, sample | sample, theoretical

e + stat_sum() x, y, size | n, prop

e + stat_summary(fun.data = "mean_cl_boot"**)**

h + stat_summary_bin(fun = "mean", geom = "bar"**)**

e + stat_identity()

e + stat_unique()

Scales Override defaults with **scales** package.

Scales map data values to the visual values of an aesthetic. To change a mapping, add a new scale.

n <– d + geom_bar(aes(fill = fl))

n + scale_fill_manual(
values = c("skyblue", "royalblue", "blue", "navy"),
limits = c("d", "e", "p", "r"), breaks =c("d", "e", "p", "r"),
name = "fuel", labels = c("D", "E", "P", "R"))

scale_ | aesthetic to adjust | prepackaged scale to use | scale-specific arguments | range of values to include in | title to use in legend/axis | labels to use in legend/axis | breaks to use in legend/axis

GENERAL PURPOSE SCALES

Use with most aesthetics

scale_*_continuous() - Map cont' values to visual ones.
scale_*_discrete() - Map discrete values to visual ones.
scale_*_binned() - Map continuous values to discrete bins.
scale_*_identity() - Use data values as visual ones.
scale_*_manual(values = c()**)** - Map discrete values to manually chosen visual ones.
scale_*_date(date_labels = "%m/%d"), date_breaks = "2 weeks"**)** - Treat data values as dates.
scale_*_datetime() - Treat data values as date times. Same as scale_*_date(). See ?strptime for label formats.

X & Y LOCATION SCALES

Use with x or y aesthetics (x shown here)

scale_x_log10() - Plot x on log10 scale.
scale_x_reverse() - Reverse the direction of the x axis.
scale_x_sqrt() - Plot x on square root scale.

COLOR AND FILL SCALES (DISCRETE)

n + scale_fill_brewer(palette = "Blues"**)**
For palette choices:
RColorBrewer::display.brewer.all()

n + scale_fill_grey(start = 0.2, end = 0.8, na.value = "red"**)**

COLOR AND FILL SCALES (CONTINUOUS)

o <– c + geom_dotplot(aes(fill = x))

o + scale_fill_distiller(palette = "Blues"**)**

o + scale_fill_gradient(low="red", high="yellow"**)**

o + scale_fill_gradient2(low = "red", high = "blue", mid = "white", midpoint = 25**)**

o + scale_fill_gradientn(colors = topo.colors(6)**)**
Also: rainbow(), heat.colors(), terrain.colors(), cm.colors(), RColorBrewer::brewer.pal()

SHAPE AND SIZE SCALES

p <– e + geom_point(aes(shape = fl, size = cyl))

p + scale_shape() + scale_size()
p + scale_shape_manual(values = c(3:7)**)**

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

p + scale_radius(range = c(1,6)**)**
p + scale_size_area(max_size = 6**)**

Coordinate Systems

r <– d + geom_bar()

r + coord_cartesian(xlim = c(0, 5)**)** - xlim, ylim
The default cartesian coordinate system.

r + coord_fixed(ratio = 1/2**)**
ratio, xlim, ylim - Cartesian coordinates with fixed aspect ratio between x and y units.

r + coord_flip()
Flip cartesian coordinates by switching x and y aesthetic mappings.

r + coord_polar(theta = "x", direction=1**)**
theta, start, direction - Polar coordinates.

r + coord_trans(y = "sqrt"**)** - x, y, xlim, ylim
Transformed cartesian coordinates. Set xtrans and ytrans to the name of a window function.

π + coord_sf() - xlim, ylim, crs. Ensures all layers use a common Coordinate Reference System.

Position Adjustments

Position adjustments determine how to arrange geoms that would otherwise occupy the same space.

s <– ggplot(mpg, aes(fl, fill = drv))

s + geom_bar(position = "dodge")
Arrange elements side by side.

s + geom_bar(position = "fill")
Stack elements on top of one another, normalize height.

e + geom_point(position = "jitter")
Add random noise to X and Y position of each element to avoid overplotting.

e + geom_label(position = "nudge")
Nudge labels away from points.

s + geom_bar(position = "stack")
Stack elements on top of one another.

Each position adjustment can be recast as a function with manual **width** and **height** arguments:
s + geom_bar(position = position_dodge(width = 1))

Themes

r + theme_bw()
White background with grid lines.

r + theme_gray()
Grey background (default theme).

r + theme_dark()
Dark for contrast.

r + theme_classic()

r + theme_light()

r + theme_linedraw()

r + theme_minimal()
Minimal theme.

r + theme_void()
Empty theme.

r + theme() Customize aspects of the theme such as axis, legend, panel, and facet properties.
r + labs(title = "Title") + theme(plot.title.position = "plot")
r + theme(panel.background = element_rect(fill = "blue"))

Faceting

Facets divide a plot into subplots based on the values of one or more discrete variables.

t <– ggplot(mpg, aes(cty, hwy)) + geom_point()

t + facet_grid(. ~ fl)
Facet into columns based on fl.

t + facet_grid(year ~ .)
Facet into rows based on year.

t + facet_grid(year ~ fl)
Facet into both rows and columns.

t + facet_wrap(~ fl)
Wrap facets into a rectangular layout.

Set **scales** to let axis limits vary across facets.

t + facet_grid(drv ~ fl, scales = "free")
x and y axis limits adjust to individual facets:
"free_x" - x axis limits adjust
"free_y" - y axis limits adjust

Set **labeller** to adjust facet label:

t + facet_grid(. ~ fl, labeller = label_both)

fl: c	fl: d	fl: e	fl: p	fl: r

t + facet_grid(fl ~ ., labeller = label_bquote(alpha ^ .(fl)**))**

α^c α^d α^e α^p α^r

Labels and Legends

Use **labs()** to label the elements of your plot.

t + labs(x = "New x axis label", **y** = "New y axis label",
title ="Add a title above the plot",
subtitle = "Add a subtitle below title",
caption = "Add a caption below plot",
alt = "Add alt text to the plot",
<AES> = "New <AES> legend title"**)**

t + annotate(geom = "text", x = 8, y = 9, label = "A"**)**
Places a geom with manually selected aesthetics.

p + guides(x = guide_axis(n.dodge = 2)**)** Avoid crowded or overlapping labels with guide_axis(n.dodge or angle).

n + guides(fill = "none"**)** Set legend type for each aesthetic: colorbar, legend, or none (no legend).

n + theme(legend.position = "bottom"**)**
Place legend at "bottom", "top", "left", or "right".

n + scale_fill_discrete(name = "Title", labels = c("A", "B", "C", "D", "E")**)**
Set legend title and labels with a scale function.

Zooming

Without clipping (preferred):

t + coord_cartesian(xlim = c(0, 100), ylim = c(10, 20)**)**

With clipping (removes unseen data points):

t + xlim(0, 100**) + ylim(**10, 20**)**

t + scale_x_continuous(limits = c(0, 100)**) + scale_y_continuous(**limits = c(0, 100)**)**

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Data import with the tidyverse : : **CHEATSHEET**

Read Tabular Data with readr

read_*(file, col_names = TRUE, col_types = NULL, col_select = NULL, id = NULL, locale, n_max = Inf, skip = 0, na = c("", "NA"), guess_max = min(1000, n_max), show_col_types = TRUE) See **?read_delim**

A|B|C
1|2|3
4|5|NA

A	B	C
1	2	3
4	5	NA

read_delim("file.txt", delim = "|"**)** Read files with any delimiter. If no delimiter is specified, it will automatically guess.
To make file.txt, run: write_file("A|B|C\n1|2|3\n4|5|NA", file = "file.txt")

A,B,C
1,2,3
4,5,NA

A	B	C
1	2	3
4	5	NA

read_csv("file.csv"**)** Read a comma delimited file with period decimal marks.
write_file("A,B,C\n1,2,3\n4,5,NA", file = "file.csv")

A;B;C
1,5;2;3
4,5;5;NA

A	B	C
1.5	2	3
4.5	5	NA

read_csv2("file2.csv"**)** Read semicolon delimited files with comma decimal marks.
write_file("A;B;C\n1,5;2;3\n4,5;5;NA", file = "file2.csv")

A B C
1 2 3
4 5 NA

A	B	C
1	2	3
4	5	NA

read_tsv("file.tsv"**)** Read a tab delimited file. Also **read_table()**.
read_fwf("file.tsv", fwf_widths(c(2, 2, NA))**)** Read a fixed width file.
write_file("A\tB\tC\n1\t2\t3\n4\t5\tNA\n", file = "file.tsv")

USEFUL READ ARGUMENTS

A	B	C
1	2	3
4	5	NA

No header
read_csv("file.csv", col_names = FALSE)

x	y	z
A	B	C
1	2	3
4	5	NA

Provide header
read_csv("file.csv",
col_names = c("x", "y", "z"))

Read multiple files into a single table
read_csv(c("f1.csv", "f2.csv", "f3.csv"),
id = "origin_file")

1	2	3
4	5	NA

Skip lines
read_csv("file.csv", skip = 1)

A	B	C
1	2	3

Read a subset of lines
read_csv("file.csv", n_max = 1)

A	B	C
NA	2	3
4	5	NA

Read values as missing
read_csv("file.csv", na = c("1"))

A;B;C
1,5;2;3,0

Specify decimal marks
read_delim("file2.csv", locale =
locale(decimal_mark = ","))

Save Data with readr

write_*(x, file, na = "NA", append, col_names, quote, escape, eol, num_threads, progress)

A	B	C
1	2	3
4	5	NA

A,B,C
1,2,3
4,5,NA

write_delim(x, file, delim = " "**)** Write files with any delimiter.

write_csv(x, file**)** Write a comma delimited file.

write_csv2(x, file**)** Write a semicolon delimited file.

write_tsv(x, file**)** Write a tab delimited file.

One of the first steps of a project is to import outside data into R. Data is often stored in tabular formats, like csv files or spreadsheets.

The front page of this sheet shows how to import and save text files into R using **readr**.

The back page shows how to import spreadsheet data from Excel files using **readxl** or Google Sheets using **googlesheets4**.

OTHER TYPES OF DATA

Try one of the following packages to import other types of files:

- **haven** - SPSS, Stata, and SAS files
- **DBI** - databases
- **jsonlite** - json
- **xml2** - XML
- **httr** - Web APIs
- **rvest** - HTML (Web Scraping)
- **readr::read_lines()** - text data

Column Specification with readr

Column specifications define what data type each column of a file will be imported as. By default readr will generate a column spec when a file is read and output a summary.

spec(x**)** Extract the full column specification for the given imported data frame.

```
spec(x)
# cols(
#   age = col_integer(),
#   edu = col_character(),
#   earn = col_double()
# )
```

age is an integer

edu is a character

earn is a double (numeric)

COLUMN TYPES

Each column type has a function and corresponding string abbreviation.

- **col_logical()** - "l"
- **col_integer()** - "i"
- **col_double()** - "d"
- **col_number()** - "n"
- **col_character()** - "c"
- **col_factor(**levels, ordered = FALSE**)** - "f"
- **col_datetime(**format = ""**)** - "T"
- **col_date(**format = ""**)** - "D"
- **col_time(**format = ""**)** - "t"
- **col_skip()** - "-", "_"
- **col_guess()** - "?"

USEFUL COLUMN ARGUMENTS

Hide col spec message
read_*(file, show_col_types = FALSE)

Select columns to import
Use names, position, or selection helpers.
read_*(file, col_select = c(age, earn))

Guess column types
To guess a column type, read_ *() looks at the first 1000 rows of data. Increase with **guess_max**.
read_*(file, guess_max = Inf)

DEFINE COLUMN SPECIFICATION

Set a default type
read_csv(
file,
col_type = list(.default = col_double()
)

Use column type or string abbreviation
read_csv(
file,
col_type = list(x = col_double(), y = "l", z = "_")
)

Use a single string of abbreviations
col types: skip, guess, integer, logical, character
read_csv(
file,
col_type = "_?ilc"
)

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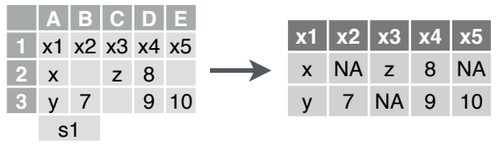
Import Spreadsheets

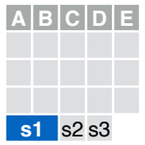
with readxl

READ EXCEL FILES



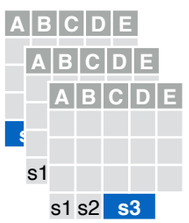
read_excel(path, sheet = NULL, range = NULL**)**
Read a .xls or .xlsx file based on the file extension. See front page for more read arguments. Also **read_xls()** and **read_xlsx()**.
read_excel("excel_file.xlsx")

READ SHEETS



read_excel(path, **sheet = NULL)** Specify which sheet to read by position or name.
read_excel(path, sheet = 1)
read_excel(path, sheet = "s1")

excel_sheets(path**)** Get a vector of sheet names.
excel_sheets("excel_file.xlsx")



To **read multiple sheets:**

1. Get a vector of sheet names from the file path.
2. Set the vector names to be the sheet names.
3. Use purrr::map() and purrr::list_rbind() to read multiple files into one data frame.

```
path <- "your_file_path.xlsx"
path |>
  excel_sheets() |>
  set_names() |>
  map(read_excel, path = path) |>
  list_rbind()
```

READXL COLUMN SPECIFICATION

Column specifications define what data type each column of a file will be imported as.

Use the **col_types** argument of **read_excel()** to set the column specification.

Guess column types
To guess a column type, read_ excel() looks at the first 1000 rows of data. Increase with the **guess_max** argument.
read_excel(path, guess_max = Inf)

Set all columns to same type, e.g. character
read_excel(path, col_types = "text")

Set each column individually
```
read_excel(
  path,
  col_types = c("text", "guess", "guess","numeric")
)
```

COLUMN TYPES

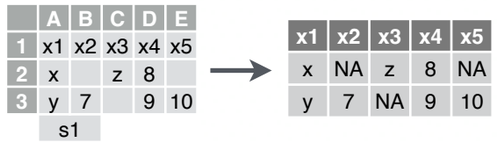
logical	numeric	text	date	list
TRUE	2	hello	1947-01-08	hello
FALSE	3.45	world	1956-10-21	1

- skip
- guess
- logical
- numeric
- text
- date
- list

Use **list** for columns that include multiple data types. See **tidyr** and **purrr** for list-column data.

with googlesheets4

READ SHEETS



read_sheet(ss, sheet = NULL, range = NULL**)**
Read a sheet from a URL, a Sheet ID, or a dribble from the googledrive package. See front page for more read arguments. Same as **range_read()**.

SHEETS METADATA

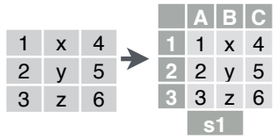
URLs are in the form:
https://docs.google.com/spreadsheets/d/**SPREADSHEET_ID**/edit#gid=**SHEET_ID**

gs4_get(ss**)** Get spreadsheet meta data.

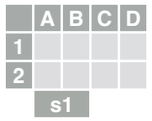
gs4_find(...**)** Get data on all spreadsheet files.

sheet_properties(ss**)** Get a tibble of properties for each worksheet. Also **sheet_names()**.

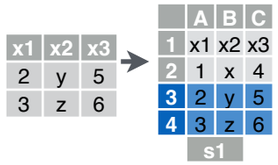
WRITE SHEETS



write_sheet(data, ss = NULL, sheet = NULL**)** Write a data frame into a new or existing Sheet.



gs4_create(name, ..., sheets = NULL**)** Create a new Sheet with a vector of names, a data frame, or a (named) list of data frames.



sheet_append(ss, data, sheet = 1**)** Add rows to the end of a worksheet.

GOOGLESHEETS4 COLUMN SPECIFICATION

Column specifications define what data type each column of a file will be imported as.

Use the **col_types** argument of **read_sheet()/ range_read()** to set the column specification.

Guess column types
To guess a column type read_sheet()/ range_read() looks at the first 1000 rows of data. Increase with **guess_max**.
read_sheet(path, guess_max = Inf)

Set all columns to same type, e.g. character
read_sheet(path, col_types = "c")

Set each column individually
# col types: skip, guess, integer, logical, character
read_sheets(ss, col_types = "_?ilc")

COLUMN TYPES

l	n	c	D	L
TRUE	2	hello	1947-01-08	hello
FALSE	3.45	world	1956-10-21	1

- skip - "_" or "-"
- guess - "?"
- logical - "l"
- integer - "i"
- double - "d"
- numeric - "n"
- date - "D"
- datetime - "T"
- character - "c"
- list-column - "L"
- cell - "C" Returns list of raw cell data.

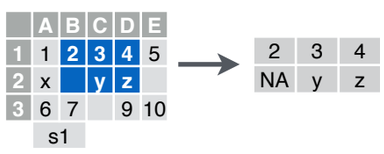
Use list for columns that include multiple data types. See **tidyr** and **purrr** for list-column data.

OTHER USEFUL EXCEL PACKAGES

For functions to write data to Excel files, see:
- **openxlsx**
- **writexl**

For working with non-tabular Excel data, see:
- **tidyxl**

CELL SPECIFICATION FOR READXL AND GOOGLESHEETS4



Use the **range** argument of **readxl::read_excel()** or **googlesheets4::read_sheet()** to read a subset of cells from a sheet.
read_excel(path, range = "Sheet1!B1:D2")
read_sheet(ss, range = "B1:D2")

Also use the range argument with cell specification functions **cell_limits()**, **cell_rows()**, **cell_cols()**, and **anchored()**.

FILE LEVEL OPERATIONS

googlesheets4 also offers ways to modify other aspects of Sheets (e.g. freeze rows, set column width, manage (work)sheets). Go to **googlesheets4.tidyverse.org** to read more.

For whole-file operations (e.g. renaming, sharing, placing within a folder), see the tidyverse package **googledrive** at **googledrive.tidyverse.org**.

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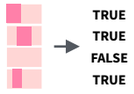
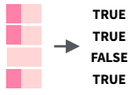
String manipulation with stringr : : **CHEATSHEET**

The **stringr** package provides a set of internally consistent tools for working with character strings, i.e. sequences of characters surrounded by quotation marks.

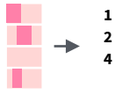
Detect Matches



str_detect(string, **pattern**, negate = FALSE**)** Detect the presence of a pattern match in a string. Also **str_like()**. str_detect(fruit, "a")



str_starts(string, **pattern**, negate = FALSE**)** Detect the presence of a pattern match at the beginning of a string. Also **str_ends()**. str_starts(fruit, "a")

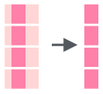


str_which(string, **pattern**, negate = FALSE**)** Find the indexes of strings that contain a pattern match. str_which(fruit, "a")

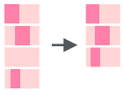
str_locate(string, **pattern**) Locate the positions of pattern matches in a string. Also **str_locate_all()**. str_locate(fruit, "a")

str_count(string, **pattern**) Count the number of matches in a string. str_count(fruit, "a")

Subset Strings



str_sub(string, start = 1L, end = -1L**)** Extract substrings from a character vector. str_sub(fruit, 1, 3); str_sub(fruit, -2)

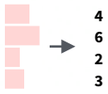


str_subset(string, **pattern**, negate = FALSE**)** Return only the strings that contain a pattern match. str_subset(fruit, "p")

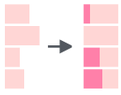
str_extract(string, **pattern**) Return the first pattern match found in each string, as a vector. Also **str_extract_all()** to return every pattern match. str_extract(fruit, "[aeiou]")

str_match(string, **pattern**) Return the first pattern match found in each string, as a matrix with a column for each ( ) group in pattern. Also **str_match_all()**. str_match(sentences, "(a|the) ([^ +])")

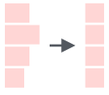
Manage Lengths



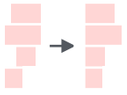
str_length(string**)** The width of strings (i.e. number of code points, which generally equals the number of characters). str_length(fruit)



str_pad(string, width, side = c("left", "right", "both"), pad = " "**)** Pad strings to constant width. str_pad(fruit, 17)



str_trunc(string, width, side = c("right", "left", "center"), ellipsis = "..."**)** Truncate the width of strings, replacing content with ellipsis. str_trunc(sentences, 6)

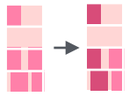


str_trim(string, side = c("both", "left", "right"))** Trim whitespace from the start and/or end of a string. str_trim(str_pad(fruit, 17))

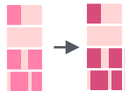
str_squish(string**)** Trim whitespace from each end and collapse multiple spaces into single spaces. str_squish(str_pad(fruit, 17, "both"))

Mutate Strings

str_sub() <- value. Replace substrings by identifying the substrings with str_sub() and assigning into the results. str_sub(fruit, 1, 3) <- "str"



str_replace(string, **pattern**, replacement**)** Replace the first matched pattern in each string. Also **str_remove()**. str_replace(fruit, "p", "-")



str_replace_all(string, **pattern**, replacement**)** Replace all matched patterns in each string. Also **str_remove_all()**. str_replace_all(fruit, "p", "-")

A STRING
↓
a string

str_to_lower(string, locale = "en"**)**[1] Convert strings to lower case. str_to_lower(sentences)

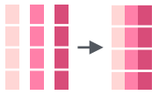
a string
↓
A STRING

str_to_upper(string, locale = "en"**)**[1] Convert strings to upper case. str_to_upper(sentences)

a string
↓
A String

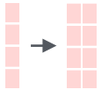
str_to_title(string, locale = "en"**)**[1] Convert strings to title case. Also **str_to_sentence()**. str_to_title(sentences)

Join and Split

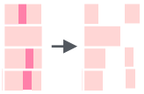


str_c(..., sep = "", collapse = NULL**)** Join multiple strings into a single string. str_c(letters, LETTERS)

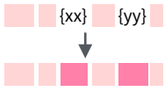
str_flatten(string, collapse = ""**)** Combines into a single string, separated by collapse. str_flatten(fruit, ", ")



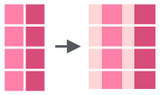
str_dup(string, times**)** Repeat strings times times. Also **str_unique()** to remove duplicates. str_dup(fruit, times = 2)



str_split_fixed(string, **pattern**, n**)** Split a vector of strings into a matrix of substrings (splitting at occurrences of a pattern match). Also **str_split()** to return a list of substrings and **str_split_i()** to return the ith substring. str_split_fixed(sentences, " ", n=3)

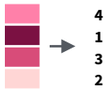


str_glue(..., .sep = "", .envir = parent.frame()**)** Create a string from strings and {expressions} to evaluate. str_glue("Pi is {pi}")



str_glue_data(.x, ..., .sep = "", .envir = parent.frame(), .na = "NA"**)** Use a data frame, list, or environment to create a string from strings and {expressions} to evaluate. str_glue_data(mtcars, "{rownames(mtcars)} has {hp} hp")

Order Strings



str_order(x, decreasing = FALSE, na_last = TRUE, locale = "en", numeric = FALSE, ...**)**[1] Return the vector of indexes that sorts a character vector. fruit[str_order(fruit)]

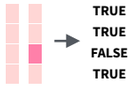
str_sort(x, decreasing = FALSE, na_last = TRUE, locale = "en", numeric = FALSE, ...**)**[1] Sort a character vector. str_sort(fruit)

Helpers

str_conv(string, encoding**)** Override the encoding of a string. str_conv(fruit,"ISO-8859-1")

appl<e>
banana
p<e>ar

str_view(string, **pattern**, match = NA**)** View HTML rendering of all regex matches. str_view(sentences, "[aeiou]")



str_equal(x, y, locale = "en", ignore_case = FALSE, ...**)**[1] Determine if two strings are equivalent. str_equal(c("a", "b"), c("a", "c"))

This is a long sentence.
↓
This is a long
sentence.

str_wrap(string, width = 80, indent = 0, exdent = 0**)** Wrap strings into nicely formatted paragraphs. str_wrap(sentences, 20)

[1] See **bit.ly/ISO639-1** for a complete list of locales.

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Need to Know

Pattern arguments in stringr are interpreted as regular expressions *after any special characters have been parsed.*

In R, you write regular expressions as *strings*, sequences of characters surrounded by quotes (**""**) or single quotes(**''**).

Some characters cannot be represented directly in an R string . These must be represented as **special characters**, sequences of characters that have a specific meaning., e.g.

Special Character	Represents
\\	\
\"	"
\n	new line

Run **?"""** to see a complete list

Because of this, whenever a \ appears in a regular expression, you must write it as \\ in the string that represents the regular expression.

Use **writeLines()** to see how R views your string after all special characters have been parsed.

```
writeLines("\\.")
# \.

writeLines("\\ is a backslash")
# \ is a backslash
```

INTERPRETATION

Patterns in stringr are interpreted as regexs. To change this default, wrap the pattern in one of:

regex(pattern, ignore_case = FALSE, multiline = FALSE, comments = FALSE, dotall = FALSE, **...)** Modifies a regex to ignore cases, match end of lines as well of end of strings, allow R comments within regex's , and/or to have . match everything including \n.
str_detect("I", regex("i", TRUE))

fixed() Matches raw bytes but will miss some characters that can be represented in multiple ways (fast). str_detect("\u0130", fixed("i"))

coll() Matches raw bytes and will use locale specific collation rules to recognize characters that can be represented in multiple ways (slow).
str_detect("\u0130", coll("i", TRUE, locale = "tr"))

boundary() Matches boundaries between characters, line_breaks, sentences, or words.
str_split(sentences, boundary("word"))

Regular Expressions -

Regular expressions, or *regexps*, are a concise language for describing patterns in strings.

MATCH CHARACTERS

see <- function(rx) str_view("abc ABC 123\t.!?\\(){}\n", rx)

string (type this)	regexp (to mean this)	matches (which matches this)	example	
	a (etc.)	a (etc.)	see("a")	abc ABC 123 .!?\(){}
\\.	\.	.	see("\\.")	abc ABC 123 .!?\(){}
\\!	\!	!	see("\\!")	abc ABC 123 .!?\(){}
\\?	\?	?	see("\\?")	abc ABC 123 .!?\(){}
\\\\	\\	\	see("\\\\")	abc ABC 123 .!?\(){}
\\(	\(	(	see("\\(")	abc ABC 123 .!?\(){}
\\)	\)	)	see("\\)")	abc ABC 123 .!?\(){}
\\{	\{	{	see("\\{")	abc ABC 123 .!?\(){}
\\}	\}	}	see("\\}")	abc ABC 123 .!?\(){}
\\n	\n	new line (return)	see("\\n")	abc ABC 123 .!?\(){}
\\t	\t	tab	see("\\t")	abc ABC 123 .!?\(){}
\\s	\s	any whitespace (**\S** *for non-whitespaces*)	see("\\s")	abc ABC 123 .!?\(){}
\\d	\d	any digit (**\D** *for non-digits*)	see("\\d")	abc ABC 123 .!?\(){}
\\w	\w	any word character (**\W** *for non-word chars*)	see("\\w")	abc ABC 123 .!?\(){}
\\b	\b	word boundaries	see("\\b")	abc ABC 123 .!?\(){}
	[:digit:][1]	digits	see("[:digit:]")	abc ABC 123 .!?\(){}
	[:alpha:][1]	letters	see("[:alpha:]")	abc ABC 123 .!?\(){}
	[:lower:][1]	lowercase letters	see("[:lower:]")	abc ABC 123 .!?\(){}
	[:upper:][1]	uppercase letters	see("[:upper:]")	abc ABC 123 .!?\(){}
	[:alnum:][1]	letters and numbers	see("[:alnum:]")	abc ABC 123 .!?\(){}
	[:punct:][1]	punctuation	see("[:punct:]")	abc ABC 123 .!?\(){}
	[:graph:][1]	letters, numbers, and punctuation	see("[:graph:]")	abc ABC 123 .!?\(){}
	[:space:][1]	space characters (i.e. \s)	see("[:space:]")	abc ABC 123 .!?\(){}
	[:blank:][1]	space and tab (but not new line)	see("[:blank:]")	abc ABC 123 .!?\(){}
	.	every character except a new line	see(".")	abc ABC 123 .!?\(){}

[1] Many base R functions require classes to be wrapped in a second set of [ ], e.g. **[[:digit:]]**

[:space:]
↵ new line

[:blank:]
space
tab

.

[:graph:]

[:punct:]
. , : ; ? ! / * @ #
- _ " ' [ ] { } ( )

[:symbol:]
| ` = + ^
~ < > $

[:alnum:]

[:digit:]
0 1 2 3 4 5 6 7 8 9

[:alpha:]

[:lower:]
a b c d e f
g h i j k l
m n o p q r
s t u v w x
y z

[:upper:]
A B C D E F
G H I J K L
M N O P Q R
S T U V W X
Y Z

ALTERNATES

alt <- function(rx) str_view("abcde", rx)

regexp	matches	example	
ab|d	or	alt("ab|d")	abcde
[abe]	one of	alt("[abe]")	abcde
[^abe]	anything but	alt("[^abe]")	abcde
[a-c]	range	alt("[a-c]")	abcde

ANCHORS

anchor <- function(rx) str_view("aaa", rx)

regexp	matches	example	
^a	start of string	anchor("^a")	aaa
a$	end of string	anchor("a$")	aaa

LOOK AROUNDS

look <- function(rx) str_view("bacad", rx)

regexp	matches	example	
a(?=c)	followed by	look("a(?=c)")	bacad
a(?!c)	not followed by	look("a(?!c)")	bacad
(?<=b)a	preceded by	look("(?<=b)a")	bacad
(?<!b)a	not preceded by	look("(?<!b)a")	bacad

QUANTIFIERS

quant <- function(rx) str_view(".a.aa.aaa", rx)

regexp	matches	example	
a?	zero or one	quant("a?")	.a.aa.aaa
a*	zero or more	quant("a*")	.a.aa.aaa
a+	one or more	quant("a+")	.a.aa.aaa
a{n}	exactly **n**	quant("a{2}")	.a.aa.aaa
a{n, }	**n** or more	quant("a{2,}")	.a.aa.aaa
a{n, m}	between **n** and **m**	quant("a{2,4}")	.a.aa.aaa

GROUPS

ref <- function(rx) str_view("abbaab", rx)

Use parentheses to set precedent (order of evaluation) and create groups

regexp	matches	example	
(ab|d)e	sets precedence	alt("(ab|d)e")	abcde

Use an escaped number to refer to and duplicate parentheses groups that occur earlier in a pattern. Refer to each group by its order of appearance

string (type this)	regexp (to mean this)	matches (which matches this)	example (the result is the same as ref("abba"))	
\\1	**\1** (etc.)	first () group, etc.	ref("(a)(b)\\2\\1")	abbaab

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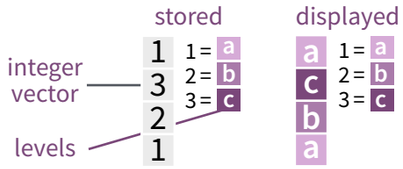
Factors with forcats : : **CHEATSHEET**

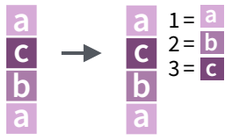
The **forcats** package provides tools for working with factors, which are R's data structure for categorical data.

Factors

R represents categorical data with factors. A **factor** is an integer vector with a **levels** attribute that stores a set of mappings between integers and categorical values. When you view a factor, R displays not the integers, but the levels associated with them.

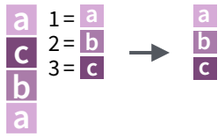


Create a factor with factor()



factor(x = character(), levels, labels = levels, exclude = NA, ordered = is.ordered(x), nmax = NA**)** Convert a vector to a factor. Also **as_factor()**.

```
f <- factor(c("a", "c", "b", "a"),
            levels = c("a", "b", "c"))
```

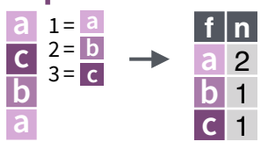


Return its levels with levels()

levels(x**)** Return/set the levels of a factor. levels(f); levels(f) <- c("x","y","z")

Use unclass() to see its structure

Inspect Factors

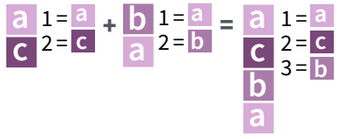


fct_count(f, sort = FALSE, prop = FALSE**)** Count the number of values with each level. fct_count(f)

fct_match(f, lvls**)** Check for lvls in f. fct_match(f, "a")

fct_unique(f**)** Return the unique values, removing duplicates. fct_unique(f)

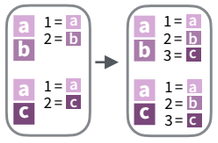
Combine Factors



fct_c(…**)** Combine factors with different levels. Also **fct_cross()**.

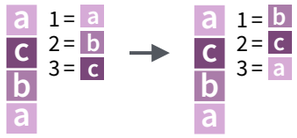
```
f1 <- factor(c("a", "c"))
f2 <- factor(c("b", "a"))
```

fct_c(f1, f2)



fct_unify(fs, levels = lvls_union(fs)**)** Standardize levels across a list of factors. fct_unify(list(f2, f1))

Change the order of levels



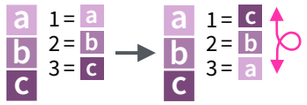
fct_relevel(.f, ..., after = 0L**)** Manually reorder factor levels. fct_relevel(f, c("b", "c", "a"))

fct_infreq(f, ordered = NA**)** Reorder levels by the frequency in which they appear in the data (highest frequency first). Also **fct_inseq()**.

```
f3 <- factor(c("c", "c", "a"))
```

fct_infreq(f3)

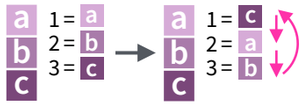
fct_inorder(f, ordered = NA**)** Reorder levels by order in which they appear in the data. fct_inorder(f2)



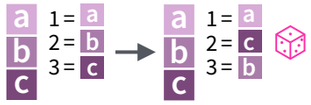
fct_rev(f**)** Reverse level order.

```
f4 <- factor(c("a","b","c"))
```

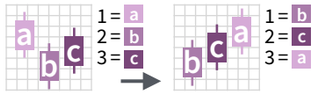
fct_rev(f4)



fct_shift(f**)** Shift levels to left or right, wrapping around end. fct_shift(f4)

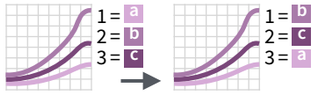


fct_shuffle(f, n = 1L**)** Randomly permute order of factor levels. fct_shuffle(f4)



fct_reorder(.f, .x, .fun = median, ..., .desc = FALSE**)** Reorder levels by their relationship with another variable.

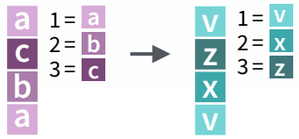
```
boxplot(
  PlantGrowth,
  weight ~ fct_reorder(group, weight)
)
```



fct_reorder2(.f, .x, .y, .fun = last2, ..., .desc = TRUE**)** Reorder levels by their final values when plotted with two other variables.

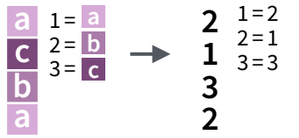
```
ggplot(
  diamonds,
  aes(
    carat, price,
    color = fct_reorder2(color, carat, price)
  )) +
  geom_smooth()
```

Change the value of levels



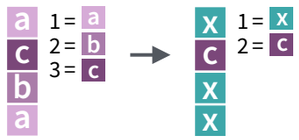
fct_recode(.f, ...**)** Manually change levels. Also **fct_relabel()** which obeys purrr::map syntax to apply a function or expression to each level.

fct_recode(f, v = "a", x = "b", z = "c")
fct_relabel(f, ~ paste0("x", .x))



fct_anon(f, prefix = ""**)** Anonymize levels with random integers.

fct_anon(f)

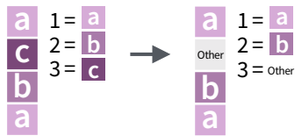


fct_collapse(.f, …, other_level = NULL**)** Collapse levels into manually defined groups.

fct_collapse(f, x = c("a", "b"))

fct_lump_min(f, min, w = NULL, other_level = "Other"**)** Lumps together factors that appear fewer than min times. Also **fct_lump_n(), fct_lump_prop(),** and **fct_lump_lowfreq()**.

fct_lump_min(f, min = 2)



fct_other(f, keep, drop, other_level = "Other"**)** Replace levels with "other."

fct_other(f, keep = c("a", "b"))

Add or drop levels

fct_drop(f, only**)** Drop unused levels.

```
f5 <- factor(c("a","b"),c("a","b","x"))
f6 <- fct_drop(f5)
```

fct_expand(f, …**)** Add levels to a factor.

fct_expand(f6, "x")

fct_na_value_to_level(f, level = "(Missing)"**)** Assigns a level to NAs to ensure they appear in plots, etc.

f7 <- factor(c("a", "b", NA))
fct_na_value_to_level(f7, level = "(Missing)")

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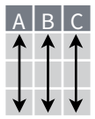
Data tidying with tidyr : : **CHEATSHEET**

Tidy data is a way to organize tabular data in a consistent data structure across packages.
A table is tidy if:

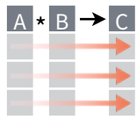


Each **variable** is in its own **column**

&

Each **observation**, or **case**, is in its own row

Access **variables** as **vectors**



Preserve **cases** in vectorized operations

Tibbles

AN ENHANCED DATA FRAME

Tibbles are a table format provided by the **tibble** package. They inherit the data frame class, but have improved behaviors:

- **Subset** a new tibble with], a vector with [[and $.
- **No partial matching** when subsetting columns.
- **Display** concise views of the data on one screen.

options(tibble.print_max = n, tibble.print_min = m, tibble.width = Inf**)** Control default display settings.

View() or **glimpse()** View the entire data set.

CONSTRUCT A TIBBLE

tibble(…**)** Construct by columns.
tibble(x = 1:3, y = c("a", "b", "c"))

tribble(…**)** Construct by rows.
tribble(~x, ~y,
1, "a",
2, "b",
3, "c")

Both make this tibble

```
A tibble: 3 × 2
      x     y
  <int> <chr>
1     1     a
2     2     b
3     3     c
```

as_tibble(x, …**)** Convert a data frame to a tibble.

enframe(x, name = "name", value = "value"**)** Convert a named vector to a tibble. Also **deframe()**.

is_tibble(x**)** Test whether x is a tibble.

Reshape Data - Pivot data to reorganize values into a new layout.

table4a

country	1999	2000
A	0.7K	2K
B	37K	80K
C	212K	213K

→

country	year	cases
A	1999	0.7K
B	1999	37K
C	1999	212K
A	2000	2K
B	2000	80K
C	2000	213K

pivot_longer(data, cols, names_to = "name", values_to = "value", values_drop_na = FALSE**)**

"Lengthen" data by collapsing several columns into two. Column names move to a new names_to column and values to a new values_to column.

pivot_longer(table4a, cols = 2:3, names_to ="year", values_to = "cases")

table2

country	year	type	count
A	1999	cases	0.7K
A	1999	pop	19M
A	2000	cases	2K
A	2000	pop	20M
B	1999	cases	37K
B	1999	pop	172M
B	2000	cases	80K
B	2000	pop	174M
C	1999	cases	212K
C	1999	pop	1T
C	2000	cases	213K
C	2000	pop	1T

→

country	year	cases	pop
A	1999	0.7K	19M
A	2000	2K	20M
B	1999	37K	172M
B	2000	80K	174M
C	1999	212K	1T
C	2000	213K	1T

pivot_wider(data, names_from = "name", values_from = "value"**)**

The inverse of pivot_longer(). "Widen" data by expanding two columns into several. One column provides the new column names, the other the values.

pivot_wider(table2, names_from = type, values_from = count)

Split Cells - Use these functions to split or combine cells into individual, isolated values.

table5

country	century	year
A	19	99
A	20	00
B	19	99
B	20	00

→

country	year
A	1999
A	2000
B	1999
B	2000

unite(data, col, …, sep = "_", remove = TRUE, na.rm = FALSE**)** Collapse cells across several columns into a single column.

unite(table5, century, year, col = "year", sep = "")

table3

country	year	rate
A	1999	0.7K/19M
A	2000	2K/20M
B	1999	37K/172M
B	2000	80K/174M

→

country	year	cases	pop
A	1999	0.7K	19M
A	2000	2K	20M
B	1999	37K	172
B	2000	80K	174

separate_wider_delim(data, cols, delim, ..., names = NULL, names_sep = NULL, names_repair = "check unique", too_few, too_many, cols_remove = TRUE) Separate each cell in a column into several columns. Also **separate_wider_regex()** and **separate_wider_position().**

separate(table3, rate, sep = "/", into = c("cases", "pop"))

table3

country	year	rate
A	1999	0.7K/19M
A	2000	2K/20M
B	1999	37K/172M
B	2000	80K/174M

→

country	year	rate
A	1999	0.7K
A	1999	19M
A	2000	2K
A	2000	20M
B	1999	37K
B	1999	172M
B	2000	80K
B	2000	174M

separate_longer_delim(data, cols, delim, .., width, keep_eampty) Separate each cell in a column into several rows.

separate_longer_delim(table3, rate, sep = "/")

Expand Tables

Create new combinations of variables or identify implicit missing values (combinations of variables not present in the data).

x

x1	x2	x3
A	1	3
B	1	4
B	2	3

→

x1	x2
A	1
A	2
B	1
B	2

expand(data, …**)** Create a new tibble with all possible combinations of the values of the variables listed in … Drop other variables.
expand(mtcars, cyl, gear, carb)

x

x1	x2	x3
A	1	3
B	1	4
B	2	3

→

x1	x2	x3
A	1	3
A	2	NA
B	1	4
B	2	3

complete(data, …, fill = list()**)** Add missing possible combinations of values of variables listed in … Fill remaining variables with NA.
complete(mtcars, cyl, gear, carb)

Handle Missing Values

Drop or replace explicit missing values (NA).

x

x1	x2
A	1
B	NA
C	NA
D	3
E	NA

→

x1	x2
A	1
D	3

drop_na(data, …**)** Drop rows containing NA's in … columns.
drop_na(x, x2)

x

x1	x2
A	1
B	NA
C	NA
D	3
E	NA

→

x1	x2
A	1
B	1
C	1
D	3
E	3

fill(data, …, .direction = "down"**)** Fill in NA's in … columns using the next or previous value.
fill(x, x2)

x

x1	x2
A	1
B	NA
C	NA
D	3
E	NA

→

x1	x2
A	1
B	2
C	2
D	3
E	2

replace_na(data, replace**)** Specify a value to replace NA in selected columns.
replace_na(x, list(x2 = 2))

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Nested Data

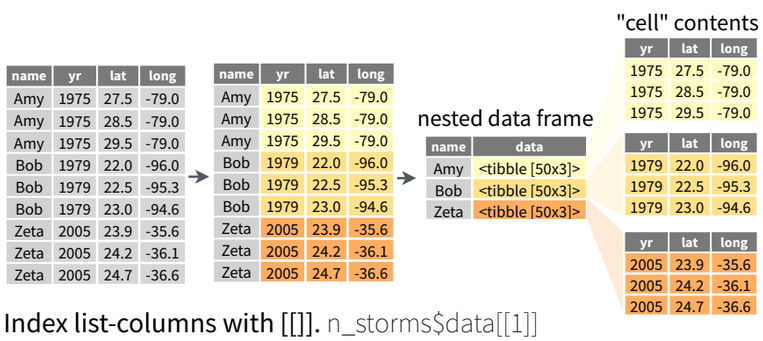
A **nested data frame** stores individual tables as a list-column of data frames within a larger organizing data frame. List-columns can also be lists of vectors or lists of varying data types. Use a nested data frame to:

- Preserve relationships between observations and subsets of data. Preserve the type of the variables being nested (factors and datetimes aren't coerced to character).
- Manipulate many sub-tables at once with **purrr** functions like map(), map2(), or pmap() or with **dplyr** rowwise() grouping.

CREATE NESTED DATA

nest(data, …**)** Moves groups of cells into a list-column of a data frame. Use alone or with dplyr::**group_by()**:

1. Group the data frame with **group_by()** and use **nest()** to move the groups into a list-column.
```
n_storms <- storms |>
  group_by(name) |>
  nest()
```
2. Use **nest(new_col = c(x, y))** to specify the columns to group using dplyr::**select()** syntax.
```
n_storms <- storms |>
  nest(data = c(year:long))
```



Index list-columns with [[]]. n_storms$data[[1]]

CREATE TIBBLES WITH LIST-COLUMNS

tibble::**tribble(**…**)** Makes list-columns when needed.
```
tribble( ~max, ~seq,
         3,    1:3,
         4,    1:4,
         5,    1:5)
```

max	seq
3	<int [3]>
4	<int [4]>
5	<int [5]>

tibble::**tibble(**…**)** Saves list input as list-columns.
```
tibble(max = c(3, 4, 5), seq = list(1:3, 1:4, 1:5))
```

tibble::**enframe(**x, name="name", value="value"**)** Converts multi-level list to a tibble with list-cols.
```
enframe(list('3'=1:3, '4'=1:4, '5'=1:5), 'max', 'seq')
```

OUTPUT LIST-COLUMNS FROM OTHER FUNCTIONS

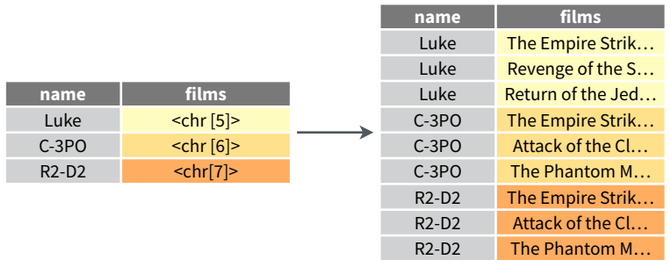
dplyr::**mutate()**, **transmute()**, and **summarise()** will output list-columns if they return a list.
```
mtcars |>
  group_by(cyl) |>
  summarise(q = list(quantile(mpg)))
```

RESHAPE NESTED DATA

unnest(data, cols, ..., keep_empty = FALSE**)** Flatten nested columns back to regular columns. The inverse of nest().
```
n_storms |> unnest(data)
```

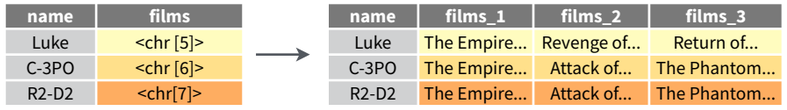
unnest_longer(data, col, values_to = NULL, indices_to = NULL**)** Turn each element of a list-column into a row.

```
starwars |>
  select(name, films) |>
  unnest_longer(films)
```



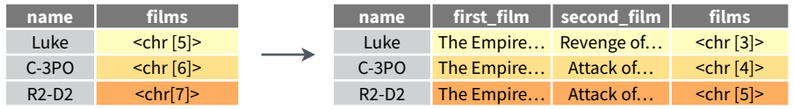
unnest_wider(data, col**)** Turn each element of a list-column into a regular column.

```
starwars |>
  select(name, films) |>
  unnest_wider(films, names_sep = "_")
```



hoist(.data, .col, ..., .remove = TRUE) Selectively pull list components out into their own top-level columns. Uses purrr::pluck() syntax for selecting from lists.

```
starwars |>
  select(name, films) |>
  hoist(films, first_film = 1, second_film = 2)
```



TRANSFORM NESTED DATA

A vectorized function takes a vector, transforms each element in parallel, and returns a vector of the same length. By themselves vectorized functions cannot work with lists, such as list-columns.

dplyr::**rowwise(**.data, …**)** Group data so that each row is one group, and within the groups, elements of list-columns appear directly (accessed with [[ ], not as lists of length one. **When you use rowwise(), dplyr functions will seem to apply functions to list-columns in a vectorized fashion.**

Apply a function to a list-column and **create a new list-column.**

```
n_storms |>
  rowwise() |>
  mutate(n = list(dim(data)))
```

dim() returns two values per row

wrap with list to tell mutate to create a list-column

Apply a function to a list-column and **create a regular column.**

```
n_storms |>
  rowwise() |>
  mutate(n = nrow(data))
```

nrow() returns one integer per row

Collapse **multiple list-columns** into a single list-column.

```
starwars |>
  rowwise() |>
  mutate(transport = list(append(vehicles, starships)))
```

append() returns a list for each row, so col type must be list

Apply a function to **multiple list-columns.**

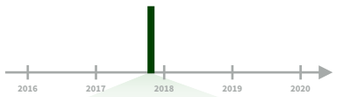
```
starwars |>
  rowwise() |>
  mutate(n_transports = length(c(vehicles, starships)))
```

length() returns one integer per row

See **purrr** package for more list functions.

Dates and times with lubridate : : **CHEATSHEET**

Date-times



2017-11-28 12:00:00

2017-11-28 12:00:00
A **date-time** is a point on the timeline, stored as the number of seconds since 1970-01-01 00:00:00 UTC

*dt <- **as_datetime(**1511870400**)***
"2017-11-28 12:00:00 UTC"

2017-11-28
A **date** is a day stored as the number of days since 1970-01-01

*d <- **as_date(**17498**)***
"2017-11-28"

12:00:00
An hms is a **time** stored as the number of seconds since 00:00:00

*t <- hms::**as_hms(**85**)***
00:01:25

PARSE DATE-TIMES (Convert strings or numbers to date-times)

1. Identify the order of the year (**y**), month (**m**), day (**d**), hour (**h**), minute (**m**) and second (**s**) elements in your data.
2. Use the function below whose name replicates the order. Each accepts a tz argument to set the time zone, e.g. ymd(x, tz = "UTC").

2017-11-28T14:02:00 — **ymd_hms(), ymd_hm(), ymd_h().** ymd_hms("2017-11-28T14:02:00")

2017-22-12 10:00:00 — **ydm_hms(), ydm_hm(), ydm_h().** ydm_hms("2017-22-12 10:00:00")

11/28/2017 1:02:03 — **mdy_hms(), mdy_hm(), mdy_h().** mdy_hms("11/28/2017 1:02:03")

1 Jan 2017 23:59:59 — **dmy_hms(), dmy_hm(), dmy_h().** dmy_hms("1 Jan 2017 23:59:59")

20170131 — **ymd(), ydm().** ymd(20170131)

July 4th, 2000 — **mdy(), myd().** mdy("July 4th, 2000")

4th of July '99 — **dmy(), dym().** dmy("4th of July '99")

2001: Q3 — **yq()** Q for quarter. yq("2001: Q3")

07-2020 — **my(), ym().** my("07-2020")

2:01 — hms::**hms()** Also lubridate::**hms(), hm()** and **ms()**, which return periods.* hms::hms(seconds = 0, minutes = 1, hours = 2)

2017.5 — **date_decimal(**decimal, tz = "UTC"**)** date_decimal(2017.5)

now(tzone = ""**)** Current time in tz (defaults to system tz). now()

today(tzone = ""**)** Current date in a tz (defaults to system tz). today()

fast_strptime() Faster strptime. fast_strptime("9/1/01", "%y/%m/%d")

parse_date_time() Easier strptime. parse_date_time("09-01-01", "ymd")

GET AND SET COMPONENTS

Use an accessor function to get a component. — d ## "2017-11-28" / day(d) ## 28

Assign into an accessor function to change a component in place. — day(d) <- 1 / d ## "2017-11-01"

2018-01-31 11:59:59 — **date(**x**)** Date component. date(dt)

2018-01-31 11:59:59 — **year(**x**)** Year. year(dt) **isoyear(**x**)** The ISO 8601 year. **epiyear(**x**)** Epidemiological year.

2018-01-31 11:59:59 — **month(**x, label, abbr**)** Month. month(dt)

2018-01-31 11:59:59 — **day(**x**)** Day of month. day(dt) **wday(**x, label, abbr**)** Day of week. **qday(**x**)** Day of quarter.

2018-01-31 11:59:59 — **hour(**x**)** Hour. hour(dt)

2018-01-31 11:59:59 — **minute(**x**)** Minutes. minute(dt)

2018-01-31 11:59:59 — **second(**x**)** Seconds. second(dt)

2018-01-31 11:59:59 UTC — **tz(**x**)** Time zone. tz(dt)

week(x**)** Week of the year. week(dt) **isoweek()** ISO 8601 week. **epiweek()** Epidemiological week.

quarter(x**)** Quarter. quarter(dt)

semester(x, with_year = FALSE**)** Semester. semester(dt)

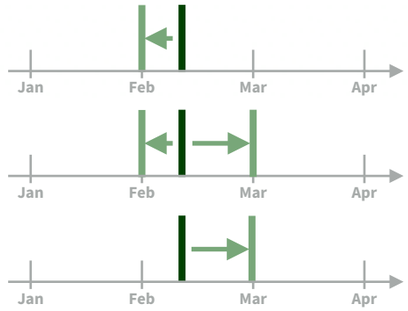
am(x**)** Is it in the am? am(dt) **pm(**x**)** Is it in the pm? pm(dt)

dst(x**)** Is it daylight savings? dst(d)

leap_year(x**)** Is it a leap year? leap_year(d)

update(object, ..., simple = FALSE**)** update(dt, mday = 2, hour = 1)

Round Date-times



floor_date(x, unit = "second"**)** Round down to nearest unit. floor_date(dt, unit = "month")

round_date(x, unit = "second"**)** Round to nearest unit. *round_date(dt, unit = "month")*

ceiling_date(x, unit = "second", change_on_boundary = NULL**)** Round up to nearest unit. ceiling_date(dt, unit = "month")

Valid units are second, minute, hour, day, week, month, bimonth, quarter, season, halfyear and year.

rollback(dates, roll_to_first = FALSE, preserve_hms = TRUE**)** Roll back to last day of previous month. Also **rollforward().** rollback(dt)

Stamp Date-times

stamp() Derive a template from an example string and return a new function that will apply the template to date-times. Also **stamp_date()** and **stamp_time()**.

1. Derive a template, create a function
   sf <- stamp("Created Sunday, Jan 17, 1999 3:34")
2. Apply the template to dates
   sf(ymd("2010-04-05"))
   ## [1] "Created Monday, Apr 05, 2010 00:00"

Tip: use a date with day > 12

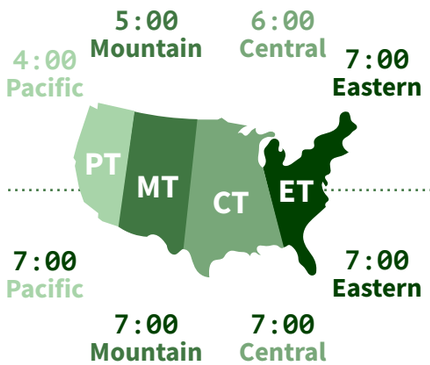
Time Zones

R recognizes ~600 time zones. Each encodes the time zone, Daylight Savings Time, and historical calendar variations for an area. R assigns *one* time zone per vector.

Use the **UTC** time zone to avoid Daylight Savings.

OlsonNames() Returns a list of valid time zone names. OlsonNames()

Sys.timezone() Gets current time zone.



with_tz(time, tzone = ""**)** Get the **same date-time** in a new time zone (a new clock time). Also **local_time(**dt, tz, units**)**. with_tz(dt, "US/Pacific")

force_tz(time, tzone = ""**)** Get the **same clock time** in a new time zone (a new date-time). Also **force_tzs()**. force_tz(dt, "US/Pacific")

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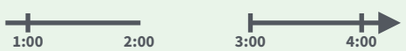
Math with Date-times – Lubridate provides three classes of timespans to facilitate math with dates and date-times.

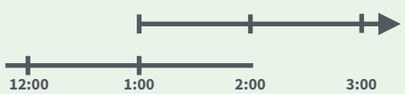
Math with date-times relies on the **timeline**, which behaves inconsistently. Consider how the timeline behaves during:

A normal day
nor <- ymd_hms("2018-01-01 01:30:00",tz="US/Eastern")

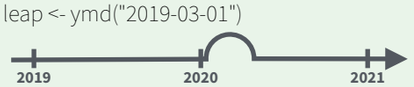
The start of daylight savings (spring forward)
gap <- ymd_hms("2018-03-11 01:30:00",tz="US/Eastern")



The end of daylight savings (fall back)
lap <- ymd_hms("2018-11-04 00:30:00",tz="US/Eastern")

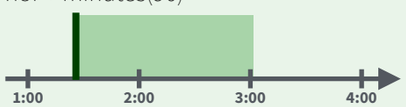


Leap years and leap seconds
leap <- ymd("2019-03-01")

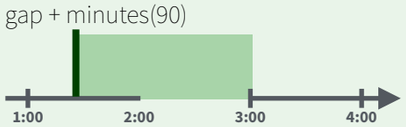


Periods track changes in clock times, which ignore time line irregularities.

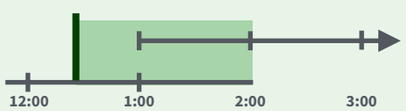
nor + minutes(90)



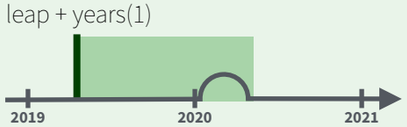
gap + minutes(90)



lap + minutes(90)

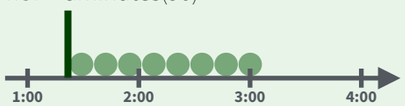


leap + years(1)

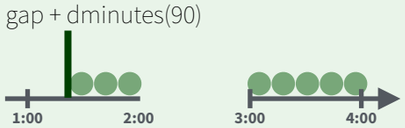


Durations track the passage of physical time, which deviates from clock time when irregularities occur.

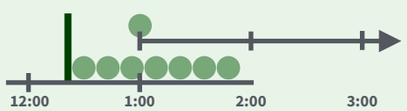
nor + dminutes(90)



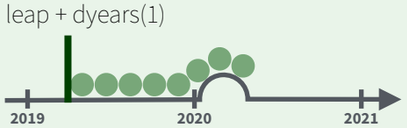
gap + dminutes(90)



lap + dminutes(90)

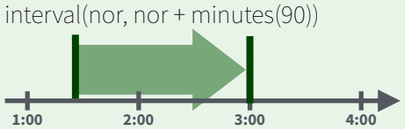


leap + dyears(1)

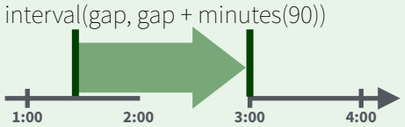


Intervals represent specific intervals of the timeline, bounded by start and end date-times.

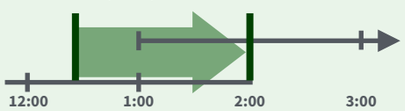
interval(nor, nor + minutes(90))



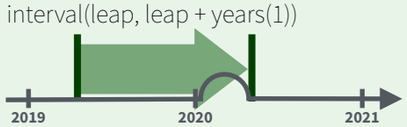
interval(gap, gap + minutes(90))



interval(lap, lap + minutes(90))



interval(leap, leap + years(1))



Not all years are 365 days due to **leap days**.

Not all minutes are 60 seconds due to **leap seconds**.

It is possible to create an imaginary date by adding **months**, e.g. February 31st

jan31 <- ymd(20180131)
jan31 + months(1)
NA

%m+% and **%m-%** will roll imaginary dates to the last day of the previous month.

jan31 %m+% months(1)
"2018-02-28"

add_with_rollback(e1, e2, roll_to_first = TRUE**)** will roll imaginary dates to the first day of the new month.

add_with_rollback(jan31, months(1), roll_to_first = TRUE)
"2018-03-01"

PERIODS

Add or subtract periods to model events that happen at specific clock times, like the NYSE opening bell.

Make a period with the name of a time unit ***pluralized***, e.g.

p <- months(3) + days(12)
p
"3m 12d 0H 0M 0S"

Number of months | Number of days | etc.

years(x = 1**)** x years.
months(x**)** x months.
weeks(x = 1**)** x weeks.
days(x = 1**)** x days.
hours(x = 1**)** x hours.
minutes(x = 1**)** x minutes.
seconds(x = 1**)** x seconds.
milliseconds(x = 1**)** x milliseconds.
microseconds(x = 1**)** x microseconds
nanoseconds(x = 1**)** x nanoseconds.
picoseconds(x = 1**)** x picoseconds.

period(num = NULL, units = "second", ...**)** An automation friendly period constructor. period(5, unit = "years")

as.period(x, unit**)** Coerce a timespan to a period, optionally in the specified units. Also **is.period()**. as.period(p)

period_to_seconds(x**)** Convert a period to the "standard" number of seconds implied by the period. Also **seconds_to_period()**. period_to_seconds(p)

DURATIONS

Add or subtract durations to model physical processes, like battery life. Durations are stored as seconds, the only time unit with a consistent length. **Difftimes** are a class of durations found in base R.

Make a duration with the name of a period prefixed with a ***d***, e.g.

dd <- ddays(14)
dd
"1209600s (~2 weeks)"

Exact length in seconds | Equivalent in common units

dyears(x = 1**)** 31536000x seconds.
dmonths(x = 1**)** 2629800x seconds.
dweeks(x = 1**)** 604800x seconds.
ddays(x = 1**)** 86400x seconds.
dhours(x = 1**)** 3600x seconds.
dminutes(x = 1**)** 60x seconds.
dseconds(x = 1**)** x seconds.
dmilliseconds(x = 1**)** x × 10^{-3} seconds.
dmicroseconds(x = 1**)** x × 10^{-6} seconds.
dnanoseconds(x = 1**)** x × 10^{-9} seconds.
dpicoseconds(x = 1**)** x × 10^{-12} seconds.

duration(num = NULL, units = "second", …**)** An automation friendly duration constructor. duration(5, unit = "years")

as.duration(x, …**)** Coerce a timespan to a duration. Also **is.duration()**, **is.difftime()**. as.duration(i)

make_difftime(x**)** Make difftime with the specified number of units. make_difftime(99999)

INTERVALS

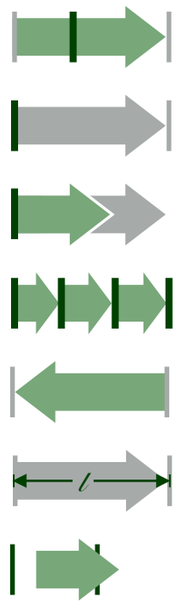
Divide an interval by a duration to determine its physical length, divide an interval by a period to determine its implied length in clock time.

Make an interval with **interval()** or **%--%**, e.g.

Start Date | End Date

i <- interval(ymd("2017-01-01"), d)
j <- d %--% ymd("2017-12-31")

2017-01-01 UTC--2017-11-28 UTC
2017-11-28 UTC--2017-12-31 UTC



a **%within%** b Does interval or date-time *a* fall within interval *b*? now() %within% i

int_start(int**)** Access/set the start date-time of an interval. Also **int_end()**. int_start(i) <- now(); int_start(i)

int_aligns(int1, int2**)** Do two intervals share a boundary? Also **int_overlaps()**. int_aligns(i, j)

int_diff(times**)** Make the intervals that occur between the date-times in a vector. v <-c(dt, dt + 100, dt + 1000); int_diff(v)

int_flip(int**)** Reverse the direction of an interval. Also **int_standardize()**. int_flip(i)

int_length(int**)** Length in seconds. int_length(i)

int_shift(int, by**)** Shifts an interval up or down the timeline by a timespan. int_shift(i, days(-1))

as.interval(x, start, …**)** Coerce a timespan to an interval with the start date-time. Also **is.interval()**. as.interval(days(1), start = now())

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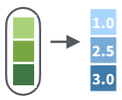
Apply functions with purrr : : **CHEATSHEET**

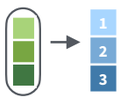
Map Functions

ONE LIST

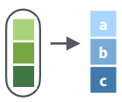
map(.x, .f, …**)** Apply a function to each element of a list or vector, and return a list.
x <- list(a = 1:10, b = 11:20, c = 21:30)
l1 <- list(x = c("a", "b"), y = c("c", "d"))
map(l1, sort, decreasing = TRUE)



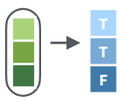
map_dbl(.x, .f, …**)**
Return a double vector.
map_dbl(x, mean)



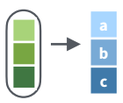
map_int(.x, .f, ...**)**
Return an integer vector.
map_int(x, length)



map_chr(.x, .f, …**)**
Return a character vector.
map_chr(l1, paste, collapse = "")



map_lgl(.x, .f, …**)**
Return a logical vector.
map_lgl(x, is.integer)



map_vec(.x, .f, ...**)**
Return a vector that is of the simplest common type.
map_vec(l1, paste, collapse = "")

walk(.x, .f, ...**)** Trigger side effects, return invisibly.
walk(x, print)

TWO LISTS

map2(.x, .y, .f, …**)** Apply a function to pairs of elements from two lists or vectors, return a list.
y <- list(1, 2, 3); z <- list(4, 5, 6); l2 <- list(x = "a", y = "z")
map2(x, y,\(x, y) x*y)

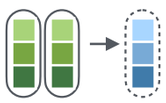
map2_dbl(.x, .y, .f, …**)** Return a double vector.
map2_dbl(y, z, ~ .x / .y)

map2_int(.x, .y, .f, …**)** Return an integer vector.
map2_int(y, z, `+`)

map2_chr(.x, .y, .f, …**)** Return a character vector.
map2_chr(l1, l2, paste, collapse = ",", sep = ":")

map2_lgl(.x, .y, .f, …**)** Return a logical vector.
map2_lgl(l2, l1, `%in%`)

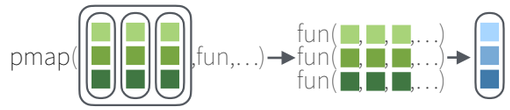
map2_vec(.x, .f, ...**)**
Return a vector that is of the simplest common type.
map2_vec(l1, l2, paste, collapse = ",", sep = ":")



walk2(.x, .y, .f, ...**)** Trigger side effects, return invisibly.
walk2(objs, paths, save)

MANY LISTS

pmap(.l, .f, …**)** Apply a function to groups of elements from a list of lists or vectors, return a list.
pmap(
list(x, y, z),
function(first, second, third) first * (second + third)
)



pmap_dbl(.l, .f, …**)**
Return a double vector.
pmap_dbl(list(y, z), ~ .x / .y)

pmap_int(.l, .f, …**)**
Return an integer vector.
pmap_int(list(y, z), `+`)

pmap_chr(.l, .f, …**)**
Return a character vector.
pmap_chr(list(l1, l2), paste, collapse = ",", sep = ":")

pmap_lgl(.l, .f, …**)**
Return a logical vector.
pmap_lgl(list(l2, l1), `%in%`)

pmap_vec(.l, .f, …**)**
Return a vector that is of the simplest common type.
pmap_vec(list(l1, l2), paste, collapse = ",", sep = ":")

pwalk(.l, .f, ...**)** Trigger side effects, return invisibly.
pwalk(list(objs, paths), save)

imap(.x, .f, ...) is shorthand for **map2(.x, names(.x), .f)** or **map2(.x, seq_along(.x), .f)** depending on whether **.x** is named or not.

Function Shortcuts

Use **\(x)** with functions like **map()** that have single arguments.

map(l, \(x) x + 2)
becomes
map(l, function(x) x + 2)

Use **\(x, y)** with functions like **map2()** that have two arguments.

map2(l, p, \(x, y) x + y)
becomes
map2(l, p, function(l, p) l + p)

Use **\(x, y, z)** etc with functions like **pmap()** that have many arguments.

pmap(list(x, y, z), \(x, y, z) x + y / z)
becomes
pmap(list(x, y, z), function(x, y, z) x * (y + z))

Use **\(x, y)** with functions like **imap()**. .x will get the list value and .y will get the index, or name if available.

imap(list("a", "b", "c"), \(x, y) paste0(y, ": ", x))
outputs **"index: value"** for each item

Use a **string** or an **integer** with any map function to index list elements by name or position. **map(l, "name")** becomes **map(l, function(x) x[["name"]])**

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Vectors

Modify

modify(.x, .f, ...**)** Apply a function to each element. Also **modify2()**, and **imodify()**.
modify(x, ~.+ 2)

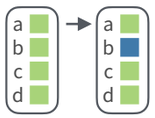

modify_at(.x, .at, .f, ...**)** Apply a function to selected elements. Also **map_at()**.
modify_at(x, "b", ~.+ 2)

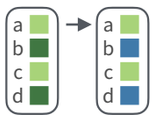

modify_if(.x, .p, .f, ...**)** Apply a function to elements that pass a test. Also **map_if()**.
modify_if(x, is.numeric,~.+2)

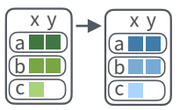

modify_depth(.x, .depth, .f, ...**)** Apply function to each element at a given level of a list. Also **map_depth()**.
modify_depth(x, 1, ~.+ 2)

Reduce

reduce(.x, .f, ..., .init, .dir = c("forward", "backward")**)**
Apply function recursively to each element of a list or vector. Also **reduce2()**.
reduce(x, sum)

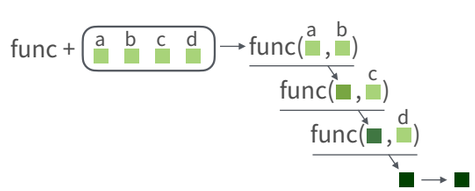

accumulate(.x, .f, ..., .init**)** Reduce a list, but also return intermediate results. Also **accumulate2()**.
accumulate(x, sum)

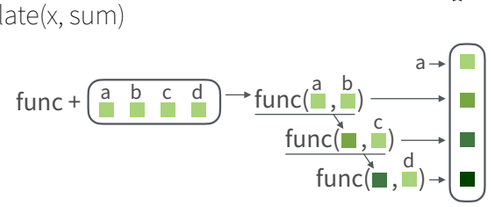

Predicate functions

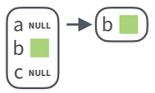

compact(.x, .p = identity**)**
Discard empty elements.
compact(x)

keep_at(x, at**)**
Keep/discard elements based by name or position. Conversely, **discard_at()**.
keep_at(x, "a")

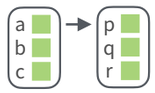

set_names(x, nm = x**)**
Set the names of a vector/list directly or with a function.
set_names(x, c("p", "q", "r"))
set_names(x, tolower)

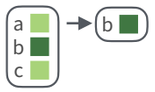

keep(.x, .p, ...**)**
Keep elements that pass a logical test. Conversely, **discard()**.
keep(x, is.numeric)

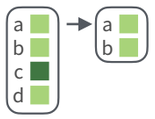

head_while(.x, .p, ...**)**
Return head elements until one does not pass. Also **tail_while()**.
head_while(x, is.character)

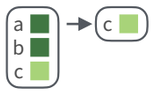

detect(.x, .f, ..., dir = c("forward", "backward"), .right = NULL, .default = NULL**)**
Find first element to pass.
detect(x, is.character)

detect_index(.x, .f, ..., dir = c("forward", "backward"), .right = NULL**)** Find index of first element to pass.
detect_index(x, is.character)

every(.x, .p, ...**)**
Do all elements pass a test?
every(x, is.character)

some(.x, .p, ...**)**
Do some elements pass a test?
some(x, is.character)

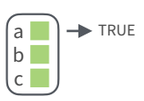

none(.x, .p, ...**)**
Do no elements pass a test?
none(x, is.character)

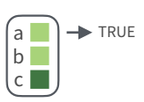

has_element(.x, .y**)**
Does a list contain an element?
has_element(x, "foo")

Pluck

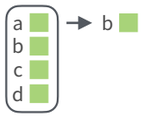

pluck(.x, ..., .default=NULL**)**
Select an element by name or index. Also **attr_getter()** and **chuck()**.
pluck(x, "b")
x |> pluck("b")

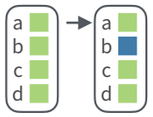

assign_in(x, where, value**)**
Assign a value to a location using pluck selection.
assign_in(x, "b", 5)
x |> assign_in("b", 5)

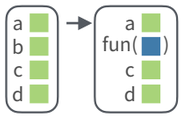

modify_in(.x, .where, .f**)** Apply a function to a value at a selected location.
modify_in(x, "b", abs)
x |> modify_in("b", abs)

Reshape

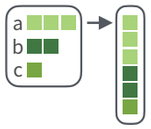

list_flatten(.x**)** Remove a level of indexes from a list.
list_flatten(x)

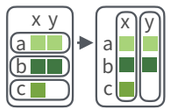

list_transpose(.l, .names = NULL**)**
Transposes the index order in a multi-level list.
list_transpose(x)

List-Columns

max	seq
3	<int [3]>
4	<int [4]>
5	<int [5]>

List-columns are columns of a data frame where each element is a list or vector instead of an atomic value. Columns can also be lists of data frames. See **tidyr** for more about nested data and list columns.

WORK WITH LIST-COLUMNS

Manipulate list-columns like any other kind of column, using **dplyr** functions like **mutate()**. Because each element is a list, use **map functions** within a column function to manipulate each element.

map(), **map2()**, or **pmap()** return lists and will **create new list-columns**.

```
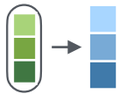
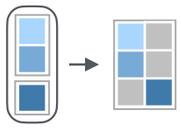
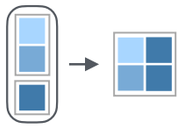
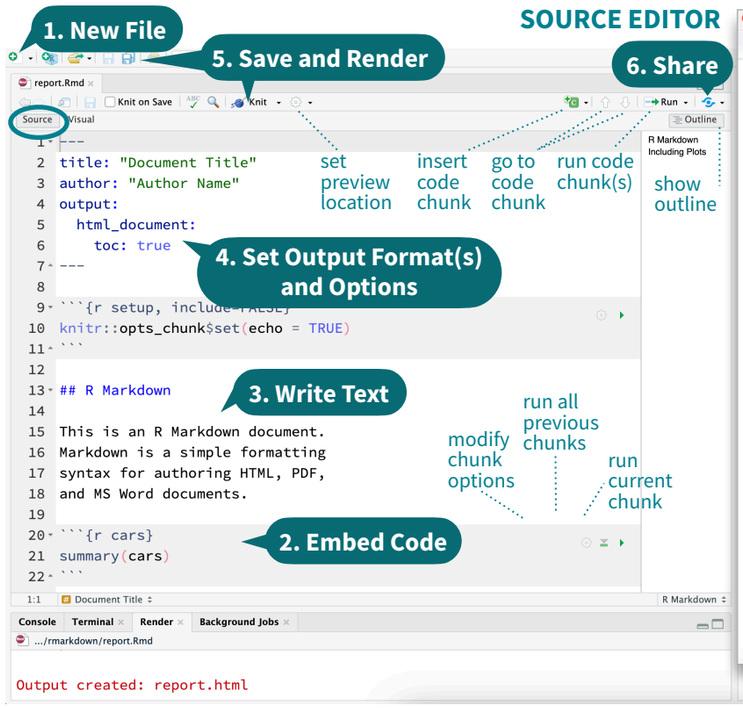
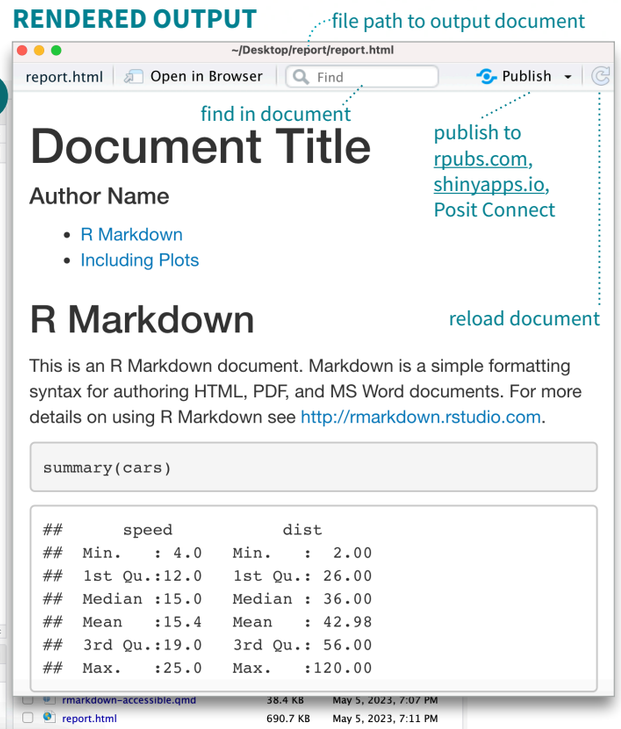
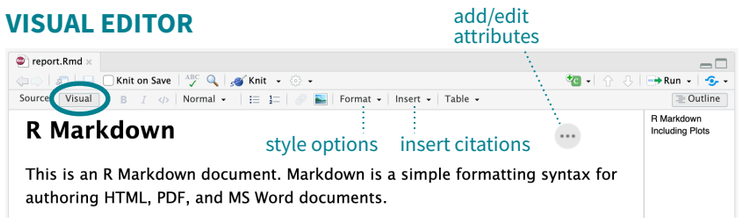
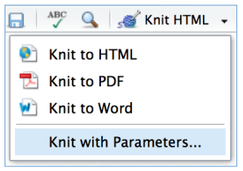
starwars |>
  transmute(ships = map2(vehicles,
                         starships,
                         append))
```

list function, return list — list-columns — column function

Suffixed map functions like **map_int()** return an atomic data type and will **simplify list-columns into regular columns**.

```
starwars |>
  mutate(n_films = map_int(films, length))
```

list function, return int — column function — list-column

Concatenate

```
x1 <- list(a = 1, b = 2, c = 3)
x2 <- list(
  a = data.frame(x = 1:2),
  b = data.frame(y = "a")
)
```



list_c(x**)** Combines elements into a vector by concatenating them together.
list_c(x1)



list_rbind(x**)** Combines elements into a data frame by row-binding them together.
list_rbind(x2)



list_cbind(x**)** Combines elements into a data frame by column-binding them together.
list_cbind(x2)

rmarkdown : : CHEATSHEET

What is rmarkdown?

.Rmd files • Develop your code and ideas side-by-side in a single document. Run code as individual chunks or as an entire document.

Dynamic Documents • Knit together plots, tables, and results with narrative text. Render to a variety of formats like HTML, PDF, MS Word, or MS Powerpoint.

Reproducible Research • Upload, link to, or attach your report to share. Anyone can read or run your code to reproduce your work.

Workflow

1. Open a **new .Rmd file** in the RStudio IDE by going to *File > New File > R Markdown*.
2. **Embed code** in chunks. Run code by line, by chunk, or all at once.
3. **Write text** and add tables, figures, images, and citations. Format with Markdown syntax or the RStudio Visual Markdown Editor.
4. **Set output format(s) and options** in the YAML header. Customize themes or add parameters to execute or add interactivity with Shiny.
5. **Save and render** the whole document. Knit periodically to preview your work as you write.
6. **Share your work!**







Embed Code with knitr

CODE CHUNKS

Surround code chunks with ```{r} and ``` or use the Insert Code Chunk button. Add a chunk label and/or chunk options inside the curly braces after **r**.

```
```{r chunk-label, include=FALSE}
summary(mtcars)
```
```

SET GLOBAL OPTIONS

Set options for the entire document in the first chunk.

```
```{r include=FALSE}
knitr::opts_chunk$set(message = FALSE)
```
```

INLINE CODE

Insert `r <code>` into text sections. Code is evaluated at render and results appear as text.

"Built with `r getRversion()`" --> "Built with 4.1.0"

OPTION	DEFAULT	EFFECTS
echo	TRUE	display code in output document
error	FALSE	TRUE (display error messages in doc) FALSE (stop render when error occurs)
eval	TRUE	run code in chunk
include	TRUE	include chunk in doc after running
message	TRUE	display code messages in document
warning	TRUE	display code warnings in document
results	"markup"	"asis" (passthrough results) "hide" (don't display results) "hold" (put all results below all code)
fig.align	"default"	"left", "right", or "center"
fig.alt	NULL	alt text for a figure
fig.cap	NULL	figure caption as a character string
fig.path	"figure/"	prefix for generating figure file paths
fig.width & fig.height	7	plot dimensions in inches
out.width		rescales output width, e.g. "75%", "300px"
collapse	FALSE	collapse all sources & output into a single block
comment	"##"	prefix for each line of results
child	NULL	files(s) to knit and then include
purl	TRUE	include or exclude a code chunk when extracting source code with knitr::purl()

See more options and defaults by running **str(knitr::opts_chunk$get())**

Insert Citations

Create citations from a bibliography file, a Zotero library, or from DOI references.

BUILD YOUR BIBLIOGRAPHY

- Add BibTeX or CSL bibliographies to the YAML header.

```
---
title: "My Document"
bibliography: references.bib
link-citations: TRUE
---
```

- If Zotero is installed locally, your main library will automatically be available.
- Add citations by DOI by searching "from DOI" in the **Insert Citation** dialog.

INSERT CITATIONS

- Access the **Insert Citations** dialog in the Visual Editor by clicking the **@** symbol in the toolbar or by clicking **Insert > Citation**.
- Add citations with markdown syntax by typing **[@cite]** or **@cite**.

Insert Tables

Output data frames as tables using **kable**(data, caption).

```
```{r}
data <- faithful[1:4, ]
knitr::kable(data,
             caption = "Table with kable")
```
```

Table with kable

eruptions	waiting
3.600	79
1.800	54
3.333	74
2.283	62

Other table packages include **flextable**, **gt**, and **kableExtra**.

Write with Markdown

The syntax on the left renders as the output on the right.

Plain text. End a line with two spaces to start a new paragraph. Also end with a backslash\ to make a new line.	Plain text. End a line with two spaces to start a new paragraph. Also end with a backslash to make a new line.
italics and **bold** superscript^2^/subscript~2~ ~~strikethrough~~ escaped: * _ \\ endash: --, emdash: ---	*italics* and **bold** superscript^2/subscript$_2$ ~~strikethrough~~ escaped: * _ \ endash: –, emdash: —
# Header 1 ## Header 2 ... ###### Header 6	**Header 1** **Header 2** ... **Header 6**

- unordered list
- item 2
  - item 2a (indent 1 tab)
  - item 2b
	• unordered list
	• item 2
	  • item 2a (indent 1 tab)
	  • item 2b

1. ordered list
2. item 2
  - item 2a (indent 1 tab)
  - item 2b
	1. ordered list
	2. item 2
	  • item 2a (indent 1 tab)
	  • item 2b

<link url> [This is a link.](link url) [This is another link][id]. At the end of the document: [id]: link url	http://www.posit.co/ This is a link. This is another link.
![Caption](image.png) or ![Caption][id2] At the end of the document: [id2]: image.png	Caption.

`verbatim code`
``
multiple lines of verbatim code
``
	`verbatim code`
	multiple lines of verbatim code

> block quotes	block quotes
equation: $e^{i \pi} + 1 = 0$ equation block: $$E = mc^{2}$$	equation: $e^{i\pi} + 1 = 0$ equation block: $E = mc^2$
horizontal rule: ---	horizontal rule:
| Right | Left | Default | Center | |---:|:---|---|:---:| | 12 | 12 | 12 | 12 | | 123 | 123 | 123 | 123 | | 1 | 1 | 1 | 1 |	Right Left Default Center 12 12 12 12 123 123 123 123 1 1 1 1

HTML Tabsets

```
## Results {.tabset}
### Plots
text

### Tables
more text
```

Results

Plots | Tables

text

Set Output Formats and their Options in YAML

Use the document's YAML header to set an **output format** and customize it with **output options**.

```
---
title: "My Document"
author: "Author Name"
output:
  html_document:
    toc: TRUE
---
```

Indent format 2 characters, indent options 4 characters

OUTPUT FORMAT	CREATES
html_document	.html
pdf_document*	.pdf
word_document	Microsoft Word (.docx)
powerpoint_presentation	Microsoft Powerpoint (.pptx)
odt_document	OpenDocument Text
rtf_document	Rich Text Format
md_document	Markdown
github_document	Markdown for Github
ioslides_presentation	ioslides HTML slides
slidy_presentation	Slidy HTML slides
beamer_presentation*	Beamer slides

* Requires LaTeX, use **tinytex::install_tinytex()**

Also see **flexdashboard**, **bookdown**, **distill**, and **blogdown**.

IMPORTANT OPTIONS	DESCRIPTION	HTML	PDF	MS Word	MS PPT
anchor_sections	Show section anchors on mouse hover (TRUE or FALSE)	X			
citation_package	The LaTeX package to process citations ("default", "natbib", "biblatex")		X		
code_download	Give readers an option to download the .Rmd source code (TRUE or FALSE)	X			
code_folding	Let readers to toggle the display of R code ("none", "hide", or "show")	X			
css	CSS or SCSS file to use to style document (e.g. "style.css")	X			
dev	Graphics device to use for figure output (e.g. "png", "pdf")	X	X		
df_print	Method for printing data frames ("default", "kable", "tibble", "paged")	X	X	X	X
fig_caption	Should figures be rendered with captions (TRUE or FALSE)	X	X	X	X
highlight	Syntax highlighting ("tango", "pygments", "kate", "zenburn", "textmate")	X	X	X	
includes	File of content to place in doc ("in_header", "before_body", "after_body")	X	X		
keep_md	Keep the Markdown .md file generated by knitting (TRUE or FALSE)	X	X	X	X
keep_tex	Keep the intermediate TEX file used to convert to PDF (TRUE or FALSE)		X		
latex_engine	LaTeX engine for producing PDF output ("pdflatex", "xelatex", or "lualatex")		X		
reference_docx/_doc	docx/pptx file containing styles to copy in the output (e.g. "file.docx", "file.pptx")			X	X
theme	Theme options (see Bootswatch and Custom Themes below)	X			
toc	Add a table of contents at start of document (TRUE or FALSE)	X	X	X	X
toc_depth	The lowest level of headings to add to table of contents (e.g. 2, 3)	X	X	X	X
toc_float	Float the table of contents to the left of the main document content (TRUE or FALSE)	X			

Use **?<output format>** to see all of a format's options, e.g. **?html_document**

Render

When you render a document, rmarkdown:

1. Runs the code and embeds results and text into an .md file with knitr.
2. Converts the .md file into the output format with Pandoc.

Save, then **Knit** to preview the document output. The resulting HTML/PDF/MS Word/etc. document will be created and saved in the same directory as the .Rmd file.

Use **rmarkdown::render()** to render/knit in the R console. See **?render** for available options.

Share

Publish on Posit Connect to share R Markdown documents securely, schedule automatic updates, and interact with parameters in real-time. **posit.co/products/enterprise/connect**.

More Header Options

PARAMETERS

Parameterize your documents to reuse with new inputs (e.g., data, values, etc.).

1. **Add parameters** in the header as sub-values of params.

```
---
params:
  state: "hawaii"
---
```

2. **Call parameters** in code using params$<name>.

```
```{r}
data <- df[, params$state]
summary(data)
```
```

3. **Set parameters** with Knit with Parameters or the params argument of **render()**.



REUSABLE TEMPLATES

1. **Create a new package** with a inst/rmarkdown/templates directory.
2. **Add a folder** containing **template.yaml** (below) and **skeleton.Rmd** (template contents).

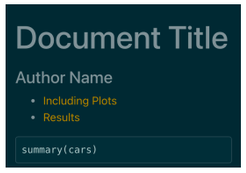
```
---
name: "My Template"
---
```

3. **Install** the package to access template by going to **File > New R Markdown > From Template**.

BOOTSWATCH THEMES

Customize HTML documents with Bootswatch themes from the **bslib** package using the theme output option.

Use **bslib::bootswatch_themes()** to list available themes.




```
---
title: "Document Title"
author: "Author Name"
output:
  html_document:
    theme:
      bootswatch: solar
---
```

CUSTOM THEMES

Customize individual HTML elements using bslib variables. Use **?bs_theme** to see more variables.

```
---
output:
  html_document:
    theme:
      bg: "#121212"
      fg: "#E4E4E4"
      base_font:
        google: "Prompt"
---
```

More on **bslib** at **pkgs.rstudio.com/bslib/**.

STYLING WITH CSS AND SCSS

Add CSS and SCSS to your document by adding a path to a file with the **css** option in the YAML header.

```
---
title: "My Document"
author: "Author Name"
output:
  html_document:
    css: "style.css"
---
```

Apply CSS styling by writing HTML tags directly or:

- Use markdown to apply style attributes inline.

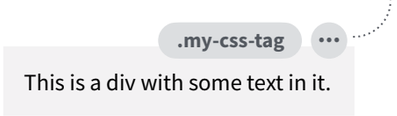
Bracketed Span
A [green]{.my-color} word.

Fenced Div
::: {.my-color}
All of these words are green.
:::

A green word.

All of these words are green.

- Use the Visual Editor. Go to **Format > Div/Span** and add CSS styling directly with Edit Attributes.



INTERACTIVITY

Turn your report into an interactive Shiny document in 4 steps:

1. Add **runtime: shiny** to the YAML header.
2. Call Shiny input functions to embed input objects.
3. Call Shiny render functions to embed reactive output.
4. Render with **rmarkdown::run()** or click **Run Document** in RStudio IDE.

```
---
output: html_document
runtime: shiny
---
```

```{r, echo = FALSE}
numericInput("n",
  "How many cars?", 5)

renderTable({
  head(cars, input$n)
})
```

How many cars? 5

	speed	dist
1	4.00	2.00
2	4.00	10.00
3	7.00	4.00
4	7.00	22.00
5	8.00	16.00

Also see Shiny Prerendered for better performance. **rmarkdown.rstudio.com/authoring_shiny_prerendered**.

Embed a complete app into your document with **shiny::shinyAppDir()**. More at **bookdown.org/yihui/rmarkdown/shiny-embedded.html**.

Shiny : : CHEATSHEET

Build an app

A **Shiny** app is a web page (**ui**) connected to a computer running a live R session (**server**).

Users can manipulate the UI, which will cause the server to update the UI's displays (by running R code).

🤖 Build with AI assistance:
gallery.shinyapps.io/assistant

💡 Get inspiration & examples:

- shiny.posit.co/r/gallery
- shinylive.io/r/examples
- `runExample()` in R console

The **UI** is a collection of input, output, and layout elements

The **server** determines how to render outputs given inputs

An **app** is a combination of UI and server logic

```
# app.R
library(shiny)

ui <- bslib::page_fluid(
  sliderInput(
    "n", "Sample size", 0, 100, 25
  ),
  plotOutput("hist")
)

server <- function(input, output) {

  output$hist <- renderPlot({
    hist(rnorm(input$n))
  })
}

shinyApp(ui, server)
```

Create `*Input()` UI and read with `input$<id>`

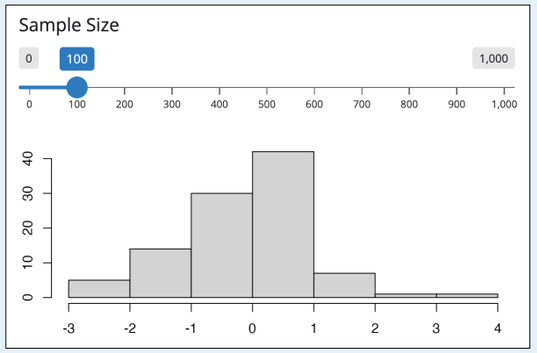
Match `*Output()` with `render*()` to add R output

Save `shinyApp()` to **app.R**

Optionally include supporting code, images, etc. in R/ and www/ folders

app-name
- app.R
- R/
- www/

Launch an **app.R** with `runApp("path/to/app-name")`.



Inputs

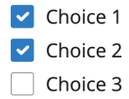
Collect values from the user.

Access the current value of an input object with `input$<id>`. Input values are **reactive**.

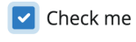
`actionButton(id, label, …)`

Action

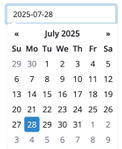
`actionLink(id, label, …)`



`checkboxGroupInput(id, label, choices, selected, …)`

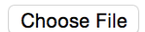


`checkboxInput(id, label, value, …)`



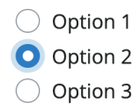
`dateInput(id, label, value, …)`

`dateRangeInput(id, label, start, end, …)`

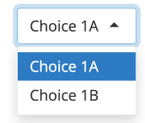


`fileInput(id, label, …)`

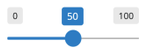
`numericInput(id, label, value, …)`



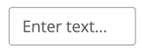
`radioButtons(id, label, choices, selected, …)`



`selectInput(id, label, choices, selected, multiple, …)`
Also `selectizeInput()`



`sliderInput(id, label, min, max, value, …)`



`textInput(id, label, value, …)`
Also `textAreaInput()`

More from the **bslib** package:

`input_dark_mode(id, mode)`

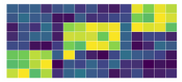
`input_switch(id, label, value, …)`

`input_task_button(id, label, value, …)`

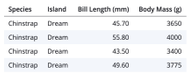
See input gallery at **shiny.posit.co/r/components**

Outputs

Reactively render R outputs



`plotOutput(id, width, height,…)`
`renderPlot(expr, …)`



Species	Island	Bill Length (mm)	Body Mass (g)
Chinstrap	Dream	45.70	3650
Chinstrap	Dream	55.80	4000
Chinstrap	Dream	43.50	3400
Chinstrap	Dream	49.60	3775

`tableOutput(id, …)`
`renderTable(expr, striped, …)`

```
  area    peri     shape perm
1 4990 2791.90 0.0903296  6.3
2 7002 3892.60 0.1486220  6.3
3 7558 3930.66 0.1833120  6.3
4 7352 3869.32 0.1170630  6.3
```

`verbatimTextOutput(id, …)`
`renderPrint(expr, …)`

Current value: 30

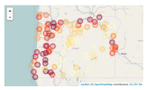
`textOutput(id, …)`
`renderText(expr, …)`

Current *value*: **30**

`uiOutput(id, …)`
`renderUI(expr, …)`

`imageOutput(id, …)`
`renderImage(expr, …)`

More from **htmlwidgets.org** ecosystem



`leafletOutput(id, …)`
`renderLeaflet(expr, …)`

`plotlyOutput(id, …)`
`renderPlotly(expr, …)`

See output gallery at **shiny.posit.co/r/components**

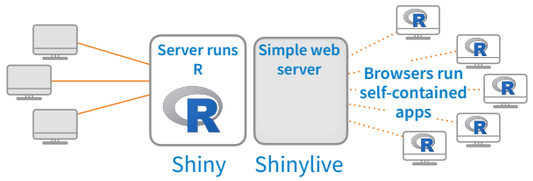
Share

Share your app in four ways:

1. **Host it on shinyapps.io**, a cloud based service from Posit. To deploy Shiny apps:
   - Create a free or professional account at **shinyapps.io**
   - Click the Publish icon in RStudio IDE, or run: `rsconnect::deployApp("path/to/app-name")`
2. **Purchase Posit Connect**, a publishing platform for R and Python. **posit.co/connect**
3. **Host your own Shiny Server** **posit.co/products/open-source/shinyserver**
4. **Export to shinylive**, a technology for running apps entirely in the browser. **posit-dev.github.io/r-shinylive**

Shinylive

Shinylive apps use WebAssembly to run entirely in a browser–no need for a server to run R.



- Edit and/or host apps at **shinylive.io/r**
- Export an app to Shinylive with `shinylive::export("app-name", "site")` Then deploy to a hosting site like Github or Netlify
- Embed Shinylive apps in Quarto sites, blogs, etc

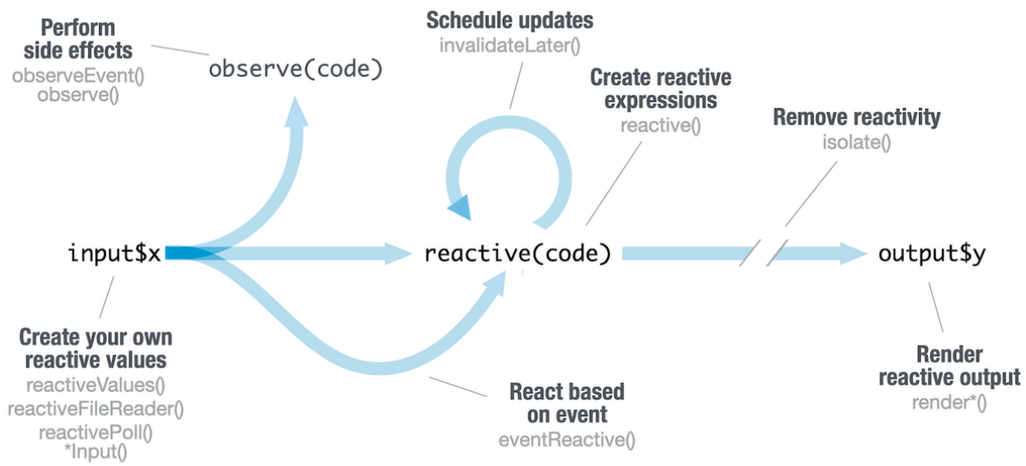
```
---
filters:
 - shinylive
---
An embedded Shinylive app:
```{shinylive-r}
#| standalone: true
# [App.py code here…]
```
```

To embed a Shinylive app in a Quarto doc, include the bold syntax.

CC BY SA Posit Software, PBC • info@posit.co • posit.co • Learn more at **shiny.posit.co** • HTML cheatsheets at **pos.it/cheatsheets** • shiny 1.11.1 • Updated: 2025-07

Reactivity

Reactive values work together with reactive functions. Call a reactive value from within the arguments of one of these functions to avoid the error **Operation not allowed without an active reactive context.**



CREATE REACTIVE VALUES

```
ui <- bslib::page_fluid(
  textInput("a", "", "A")
)

server <- \(input, output){
  print(isolate(input$a))

  rv <- reactiveVal(NULL)
  print(isolate(rv()))
}

shinyApp(ui, server)
```

`*Input()` functions

Create a reactive value `input$<id>` from user input.

`reactiveVal(value)`

Create a reactive value from a given value. Useful for managing state.

RENDER REACTIVE OUTPUT

```
ui <- bslib::page_fluid(
  textInput("a", "", "A"),
  textOutput("b")
)

server <- \(input, output){
 output$b <- renderText({
   input$a
  })
}

shinyApp(ui, server)
```

`render*()` functions

Produces results for a corresponding `*Output()` UI container. Re-render occurs when reactive dependencies change.

Save the results to `output$<id>`.

CREATE REACTIVE EXPRESSIONS

```
ui <- bslib::page_fluid(
 textInput("a", "", "A"),
 textInput("z", "", "Z"),
 textOutput("b"))

server <- \(input, output){
 re <- reactive({
  paste(input$a, input$z)
 })
 output$b <- renderText({
  re()
 })
}

shinyApp(ui, server)
```

`reactive(x)`

Calculate a (reactive) value based on other reactive values.

Useful for encapsulating reactive logic needed across multiple outputs.

PERFORM SIDE EFFECTS

```
ui <- bslib::page_fluid(
  textInput("a", "", "A"),
  actionButton("go", "Go")
)

server <- \(input, output){
  observe(print(input$a))

  observeEvent(input$go, {
    print(input$a)
  })
}

shinyApp(ui, server)
```

`observe(x)`

Observe changes to reactive values

`observeEvent(eventExpr, handlerExpr)`

Runs code in 2nd argument when 1st argument changes.

REACT BASED ON EVENT

```
ui <- bslib::page_fluid(
 textInput("a", "", "A"),
 actionButton("go", "Go"),
 textOutput("b")
)

server <- \(input, output){
 re <- eventReactive(
  input$go, {input$a}
 )
 output$b <- renderText({
  re()
 })
}

shinyApp(ui, server)
```

`eventReactive(eventExpr, valueExpr)`

Creates reactive expression with code in 2nd argument that only invalidates when reactive values in 1st argument change.

REMOVE REACTIVE DEPENDENCIES

```
ui <- bslib::page_fluid(
  textInput("a", "", "A"),
  actionButton("go", "Go"),
  textOutput("b")
)

server <- \(input, output){

 output$b <- renderText({
   input$go
   isolate(input$a)
  })

}

shinyApp(ui, server)
```

`isolate(expr)`

Prevent reactive values from invalidating a reactive expression.

User Interfaces (UI)

Design delightful UI with the **bslib** package. It provides layouts, components, themes, & more.

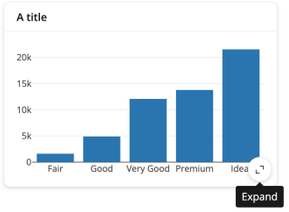
PAGE LAYOUTS

`page_sidebar()`	Screen-filling sidebar layout
`page_fillable()`	Screen-filling page layout
`page_fixed()`	Constrained width page
`page_fluid()`	Basic full-width page
`page_navbar()`	Multi-page app with a top nav bar

CARDS

Visually group UI elements together with the `card()` component.

```
card(full_screen = T,
  card_header("A title"),
  plotOutput("my_output"),
  card_footer("A footer")
)
```



UI LAYOUTS

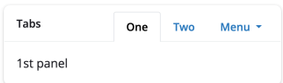
Multiple columns

`layout_columns()`	Bootstrap's 12-column grid
`layout_column_wrap()`	Equal-width columns
`layout_sidebar()`	Resizable 2-column layout

Multiple panels

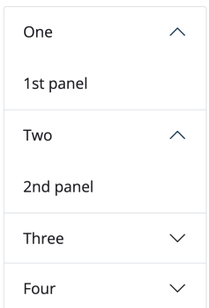
```
navset_card_underline(
  nav_panel(
    "One", "1st panel"),
  nav_panel(
    "Two", "2nd panel"),
  nav_menu("Menu",
    nav_panel("3", "3rd"))
)
```

Navigate a set of `nav_panel()`s in various ways with `navset_card_*`



ACCORDIONS

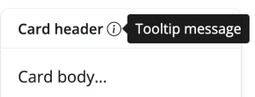
```
accordion(
 open = c("One", "Two"),
 accordion_panel(
   "One", "1st panel"),
 accordion_panel(
   "Two", "2nd panel"),
 accordion_panel(
   "Three", "3rd panel"),
)
```



Tip: place within `sidebar()` to group similar inputs

TOOLTIPS

```
tooltip(
  icon("info-circle"),
  "Tooltip message"
)
```



VALUE BOXES

```
value_box(
  "Title",
  "Value",
  showcase=icon()
)
```

Custom UI

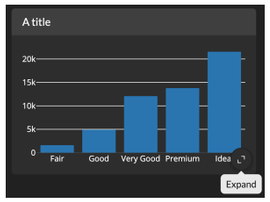
Custom how your app looks and behaves.

THEMES

Bootswatch

Choose from over a dozen pre-packaged themes

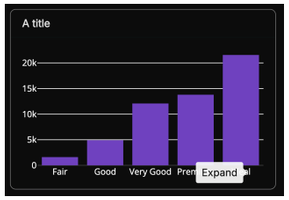
```
library(bslib)
theme <- bs_theme(
  bootswatch="darkly"
)
ui <- page_fluid(
  theme=theme, ...
)
```



Custom themes

Quickly change main colors and fonts. Change in real-time by adding **bs_themer()** to your UI.

```
bs_theme(
  bg = "#222",
  fg = "white",
  primary = "purple",
  base_font =
font_google("Inter")
)
```



CUSTOM HTML

Shiny UI is powered by HTML, CSS, and JS:

```
page_fluid(class = "pt-3")
#> <div class="container-fluid pt-3"></div>
```

If you know these web technologies, you can customize UI to your heart's content. Start small by modifying/authoring HTML and including CSS/JS snippets. Or, go fully custom with `htmlTemplate()`

HTML Add HTML elements with `tags`, a list of functions that parallel common HTML tags, e.g. `tags$a()`. Unnamed arguments are treated as children and named arguments become HTML attributes.

CSS To include a CSS file, use `includeCSS()`, or
1. Place the file in the `www` subdirectory
2. Link to it with:

```
tags$head(tags$link(href="<file name>",
rel="stylesheet"))
```

JS To include JS, use `includeScript()` or
1. Place the file in the `www` subdirectory
2. Link to it with:

```
tags$head(tags$script(src="<file name>"))
```

IMAGE To include an image:
1. Place the file in the `www` subdirectory
2. Link to it with `img(src="<file name>")`